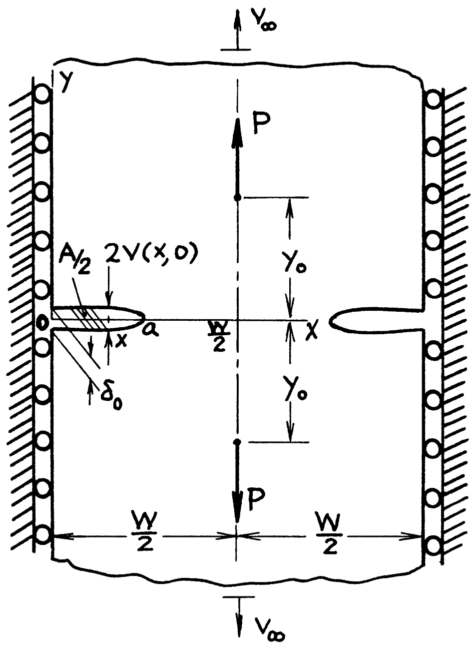


$$K_I = \frac{P}{\sqrt{W}}\sqrt{\tan\frac{\pi a}{W}}\left(1-\alpha y_0\frac{\partial}{\partial y_0}\right)\frac{\sinh\frac{\pi y_0}{W}}{\sqrt{\left(\cos\frac{\pi a}{W}\right)^2+\left(\sinh\frac{\pi y_0}{W}\right)^2}}$$

Crack Opening Area:

$$A = \frac{8PW}{\pi E'}\left(1-\alpha y_0\frac{\partial}{\partial y_0}\right)\left\{\sinh^{-1}\left(\frac{\sinh\frac{\pi y_0}{W}}{\cos\frac{\pi a}{W}}\right)-\frac{\pi y_0}{W}\right\}$$

Relative Displacement at Infinity:

$$2\,v_\infty = {}^{A}/_{W}$$

Crack Opening Profile:

$$2\,v(x,\,0) = \frac{4P}{\pi E'}\left(1-\alpha y_0\frac{\partial}{\partial y_0}\right)\tanh^{-1}\sqrt{\frac{1-\left(\cos\frac{\pi a}{W}/\cos\frac{\pi x}{W}\right)^2}{1+\left(\cos\frac{\pi a}{W}/\sinh\frac{\pi y_0}{W}\right)^2}}$$

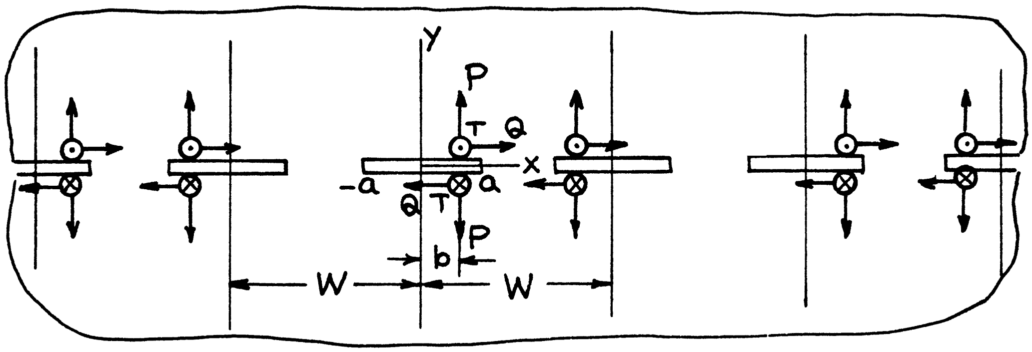
Opening at Edge:

$$\delta_0 = 2\,v(0,\,0) = \frac{4P}{\pi E'}\left(1-\alpha y_0\frac{\partial}{\partial y_0}\right)\tanh^{-1}\left\{\frac{\sin\frac{\pi a}{W}}{\sqrt{1+\left(\cos\frac{\pi a}{W}/\sinh\frac{\pi y_0}{W}\right)^2}}\right\}$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Methods: Westergaard Stress Function, Reciprocity (see **page 7.1a**)
Accuracy: Exact
Reference: **Tada 1985**



$$z = x + iy$$

$$\left\{\begin{matrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{matrix}\right\} = \frac{1}{W}\left\{\begin{matrix} P \\ Q \\ T \end{matrix}\right\} \frac{\cos\frac{\pi b}{W}\sqrt{\left(\sin\frac{\pi a}{W}\right)^2 - \left(\sin\frac{\pi b}{W}\right)^2}}{\left(\sin\frac{\pi z}{W} - \sin\frac{\pi b}{W}\right)\sqrt{\left(\sin\frac{\pi z}{W}\right)^2 - \left(\sin\frac{\pi a}{W}\right)^2}}$$

$$\left\{\begin{matrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{matrix}\right\} = \frac{1}{\pi}\left\{\begin{matrix} P \\ Q \\ T \end{matrix}\right\} \left[\tan^{-1}\sqrt{\frac{\left(\cos\frac{\pi a}{W} / \cos\frac{\pi z}{W}\right)^2 - 1}{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi b}{W}\right)^2}}\right.$$

$$\left. + i\ \cot\frac{\pi b}{W}\sqrt{1 - \left(\sin\frac{\pi b}{W} / \sin\frac{\pi a}{W}\right)^2}\ \Pi_c\left(\sin^{-1}\left(\frac{\sin\frac{\pi z}{W}}{\sin\frac{\pi a}{W}}\right), \left(\frac{\sin\frac{\pi a}{W}}{\sin\frac{\pi b}{W}}\right)^2, \ \sin\frac{\pi a}{W}\right)\right]$$

$$\left\{\begin{matrix} K_I \\ K_{II} \\ K_{III} \end{matrix}\right\}_{\pm a} = \frac{1}{W}\left\{\begin{matrix} P \\ Q \\ T \end{matrix}\right\}\sqrt{W\tan\frac{\pi a}{W}}\ \frac{\cos\frac{\pi b}{W}}{\sin\frac{\pi a}{W}}\sqrt{\frac{\sin\frac{\pi a}{W} \pm \sin\frac{\pi b}{W}}{\sin\frac{\pi a}{W} \mp \sin\frac{\pi b}{W}}}$$

$$\operatorname{Im}\left\{\begin{matrix}\overline{Z}_I(x)\\ \overline{Z}_{II}(x)\\ \overline{Z}_{III}(x)\end{matrix}\right\}_{|x|\le a} = \frac{1}{\pi}\left\{\begin{matrix}P\\ Q\\ T\end{matrix}\right\}\left[\begin{pmatrix}\tanh^{-1}\\ \coth^{-1}\end{pmatrix}\sqrt{\frac{1-\left(\cos\frac{\pi a}{W}/\cos\frac{\pi b}{W}\right)^2}{1-\left(\cos\frac{\pi a}{W}/\cos\frac{\pi x}{W}\right)^2}}\quad\begin{pmatrix}-a<x<b\\ b<x<a\end{pmatrix}\right.$$

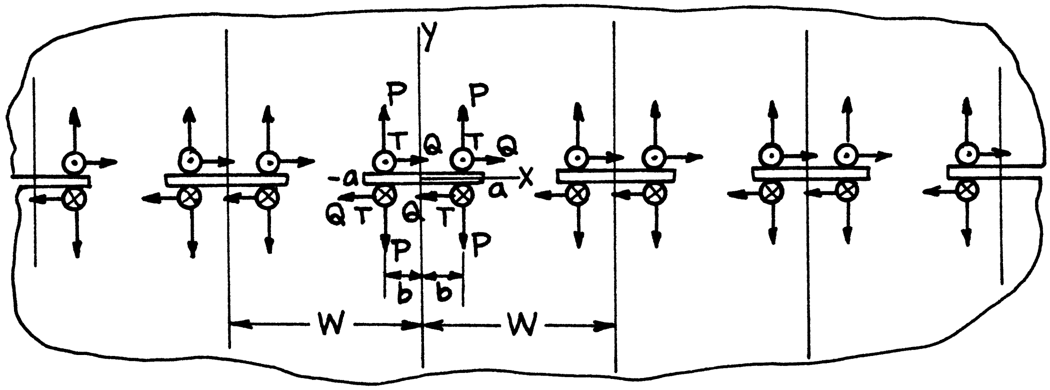
$$\left. + \cot\frac{\pi b}{W}\sqrt{1-\left(\sin\frac{\pi b}{W}/\sin\frac{\pi a}{W}\right)^2}\,\Pi\left(\sin^{-1}\left(\frac{\sin\frac{\pi x}{W}}{\sin\frac{\pi a}{W}}\right),\left(\frac{\sin\frac{\pi a}{W}}{\sin\frac{\pi b}{W}}\right)^2,\ \sin\frac{\pi a}{W}\right)\right]$$

where

$$\Pi_c(\varphi, n, k) = \int \frac{d\varphi}{\left(1-n\sin^2\varphi\right)\sqrt{1-k^2\sin^2\varphi}}$$

$$\Pi(\varphi, n, k) = \int_0^{\varphi} \frac{d\varphi}{\left(1-n\sin^2\varphi\right)\sqrt{1-k^2\sin^2\varphi}}$$

Method: Westergaard Stress Function
Accuracy: Exact
References: **Irwin 1957; Tada 1973**



$$z = x + iy$$

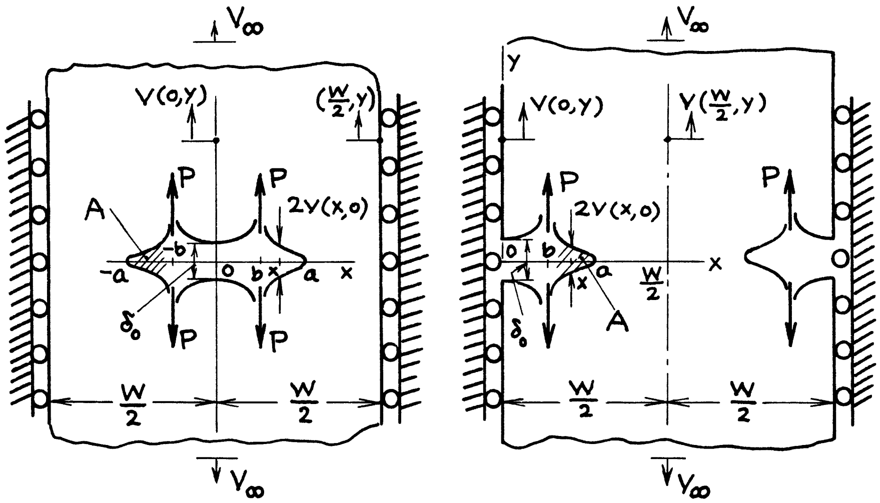
$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{2}{W} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \frac{\cos\frac{\pi b}{W}\sqrt{\left(\sin\frac{\pi a}{W}\right)^2 - \left(\sin\frac{\pi b}{W}\right)^2}}{\left\{\left(\sin\frac{\pi z}{W}\right)^2 - \left(\sin\frac{\pi b}{W}\right)^2\right\}\sqrt{1 - \left(\sin\frac{\pi a}{W} / \sin\frac{\pi z}{W}\right)^2}}$$

$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \tan^{-1} \sqrt{\frac{\left(\cos\frac{\pi a}{W} / \cos\frac{\pi z}{W}\right)^2 - 1}{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi b}{W}\right)^2}}$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{W} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \sqrt{W \tan\frac{\pi a}{W}} \frac{\cos\frac{\pi b}{W}}{\sqrt{\left(\sin\frac{\pi a}{W}\right)^2 - \left(\sin\frac{\pi b}{W}\right)^2}}$$

$$\operatorname{Im} \begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{Bmatrix}_{|x| \le a} = \frac{1}{\pi} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \ln \left| \frac{\sqrt{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi x}{W}\right)^2} + \sqrt{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi b}{W}\right)^2}}{\sqrt{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi x}{W}\right)^2} - \sqrt{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi b}{W}\right)^2}} \right|$$

Method: Westergaard Stress Function
Accuracy: Exact
References: **Irwin 1957, 1958a**



$$K_I = \frac{2P}{\sqrt{W}}\sqrt{\tan\frac{\pi a}{W}}\,\frac{\cos\frac{\pi b}{W}}{\sqrt{\left(\sin\frac{\pi a}{W}\right)^2 - \left(\sin\frac{\pi b}{W}\right)^2}}$$

Crack Opening Area:

$$A = \frac{8PW}{\pi E'}\cosh^{-1}\left(\cos\frac{\pi b}{W} \Big/ \cos\frac{\pi a}{W}\right)$$

Relative Displacement at Infinity:

$$2v_\infty = A/W$$

Crack Opening Profile:

$$\underset{|x|\le a}{2v(x,0)} = \frac{8P}{\pi E'}\begin{pmatrix}\tanh^{-1}\\ \coth^{-1}\end{pmatrix}\sqrt{\frac{1-\left(\cos\frac{\pi a}{W} / \cos\frac{\pi b}{W}\right)^2}{1-\left(\cos\frac{\pi a}{W} / \cos\frac{\pi x}{W}\right)^2}}\begin{pmatrix}|x| \le b\\ b < |x| \le a\end{pmatrix}$$

Opening at Center or Edge:

$$\delta_0 = 2v(0,0) = \frac{8P}{\pi E'}\tanh^{-1}\frac{\sqrt{1-\left(\cos\frac{\pi a}{W} / \cos\frac{\pi b}{W}\right)^2}}{\sin\frac{\pi a}{W}}$$

Vertical Displacement at $(0, y)$:

$$v(0,y) = \frac{4P}{\pi E'}\left(1 - \alpha y \frac{\partial}{\partial y}\right) \tanh^{-1} \sqrt{\frac{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi b}{W}\right)^2}{1 - \left(\cos\frac{\pi a}{W} / \cosh\frac{\pi y}{W}\right)^2}}$$

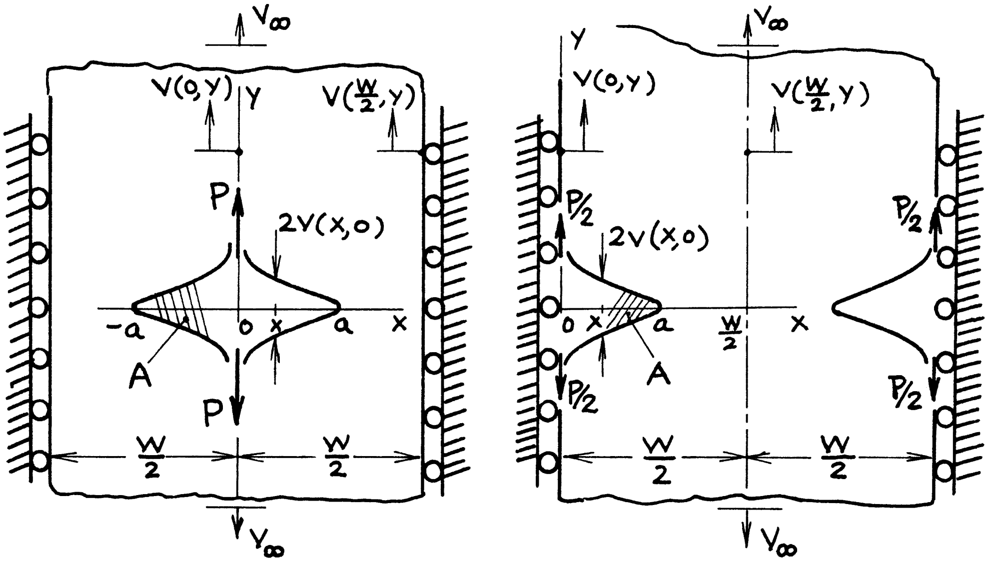
Vertical Displacement at $({}^{W}\!/_{2}, y)$:

$$v\left(\frac{W}{2}, y\right) = \frac{4P}{\pi E'}\left(1 - \alpha y \frac{\partial}{\partial y}\right) \tanh^{-1} \sqrt{\frac{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi b}{W}\right)^2}{1 + \left(\cos\frac{\pi a}{W} / \sinh\frac{\pi y}{W}\right)^2}}$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Methods: Westergaard Stress Function, Reciprocity (see **pages 7.1, 7.1a, 7.4a, 7.5a**)
Accuracy: Exact
Reference: **Tada 1985**



$$K_I = \frac{P}{\sqrt{\pi a}} \sqrt{\frac{2\pi a}{W} \Big/ \sin\frac{2\pi a}{W}}$$

Crack Opening Area:

$$A = \frac{4PW}{\pi E'} \cosh^{-1}\left(\sec\frac{\pi a}{W}\right)$$

Relative Displacement at Infinity:

$$2v_\infty = {}^{A}\!/_{W} = \frac{4P}{\pi E'} \cosh^{-1}\left(\sec\frac{\pi a}{W}\right)$$

Crack Opening Profile:

$$\underset{|x|\le a}{2v(x,0)} = \frac{4P}{\pi E'} \tanh^{-1}\left\{ \frac{\sqrt{1-\left(\cos\frac{\pi a}{W}\Big/\cos\frac{\pi x}{W}\right)^2}}{\sin\frac{\pi a}{W}} \right\}$$

Vertical Displacements at $(0, y)$ and $({}^{W}\!/_{2}, y)$:

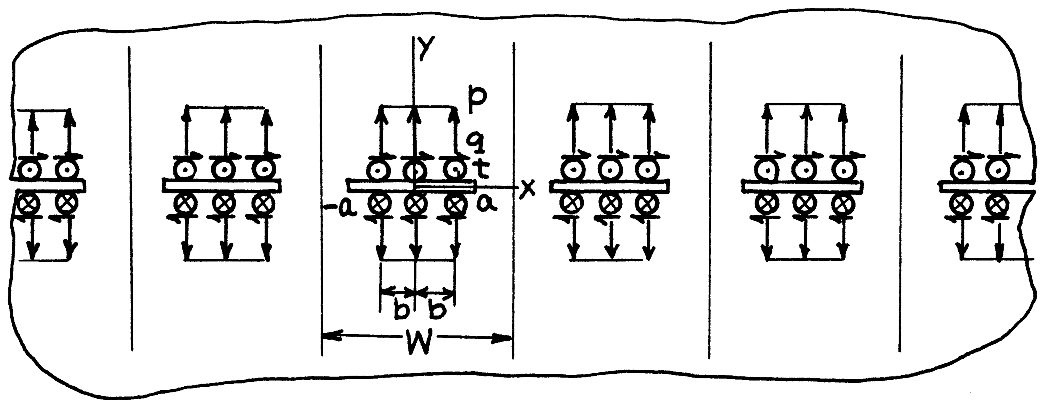
$$v(0,y) = \frac{2P}{\pi E'}\left(1-\alpha y\frac{\partial}{\partial y}\right) \tanh^{-1}\left\{ \frac{\sin\frac{\pi a}{W}}{\sqrt{1-\left(\cos\frac{\pi a}{W}\Big/\cosh\frac{\pi y}{W}\right)^2}} \right\}$$

$$v\left(\frac{W}{2}, y\right) = \frac{2P}{\pi E'}\left(1 - \alpha y \frac{\partial}{\partial y}\right) \tanh^{-1}\left\{\frac{\sin\frac{\pi a}{W}}{\sqrt{1 + \left(\cos\frac{\pi a}{W} \Big/ \sinh\frac{\pi y}{W}\right)^2}}\right\}$$

where

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Method: Special Case of **page 7.7a**
Accuracy: Exact
Reference: **Tada 1985**



$$z = x + iy$$

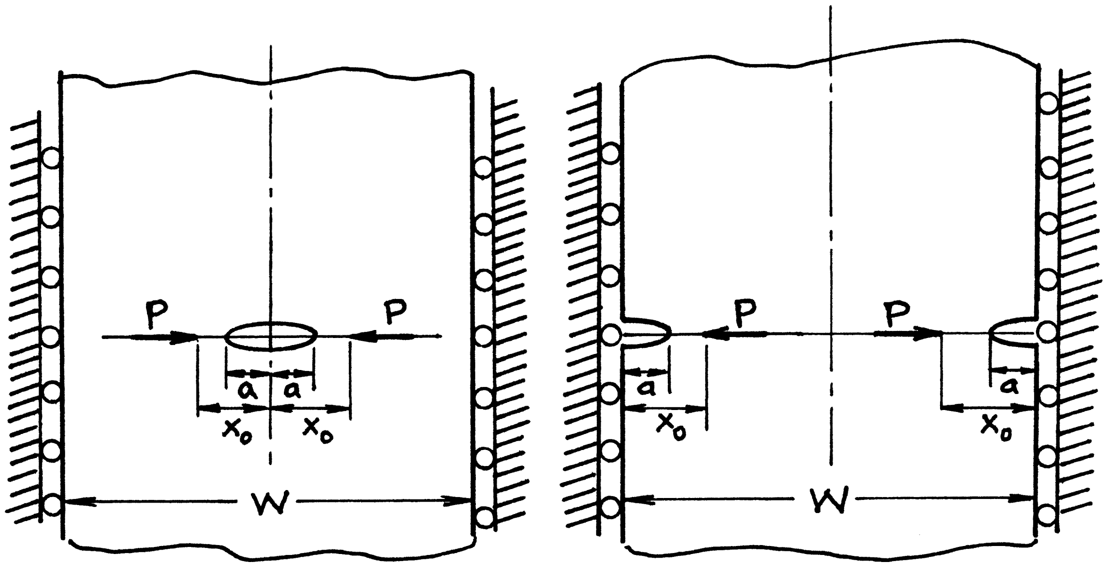
$$\begin{Bmatrix} Z_I(z) \\ Z_{II}(z) \\ Z_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \left[\frac{\sin^{-1}\left(\sin\frac{\pi b}{W} / \sin\frac{\pi a}{W}\right)}{\sqrt{1 - \left(\sin\frac{\pi a}{W} / \sin\frac{\pi z}{W}\right)^2}} + \tan^{-1} \sqrt{\frac{\left(\sin\frac{\pi a}{W} / \sin\frac{\pi b}{W}\right)^2 - 1}{1 - \left(\sin\frac{\pi a}{W} / \sin\frac{\pi z}{W}\right)^2}} \right]$$

$$\begin{Bmatrix} \overline{Z}_I(z) \\ \overline{Z}_{II}(z) \\ \overline{Z}_{III}(z) \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \int_0^b \left\{ -\tan^{-1} \sqrt{\frac{\left(\cos\frac{\pi a}{W} / \cos\frac{\pi z}{W}\right)^2 - 1}{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi x_0}{W}\right)^2}} \right\} dx_0$$

$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \sqrt{W \tan\frac{\pi a}{W}} \cdot \sin^{-1}\left(\frac{\sin\frac{\pi b}{W}}{\sin\frac{\pi a}{W}} \right)$$

$$\operatorname{Im} \begin{Bmatrix} \overline{Z}_I(x) \\ \overline{Z}_{II}(x) \\ \overline{Z}_{III}(x) \end{Bmatrix}_{|x| \le a} = \frac{1}{\pi} \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \int_0^b \ln \left| \frac{\sqrt{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi x}{W}\right)^2} + \sqrt{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi x_0}{W}\right)^2}}{\sqrt{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi x}{W}\right)^2} - \sqrt{1 - \left(\cos\frac{\pi a}{W} / \cos\frac{\pi x_0}{W}\right)^2}} \right| dx_0$$

Method: Westergaard Stress Function (Integration of **page 7.6** or **page 7.7**)
Accuracy: Exact
References: **Irwin 1957; Tada 1973**

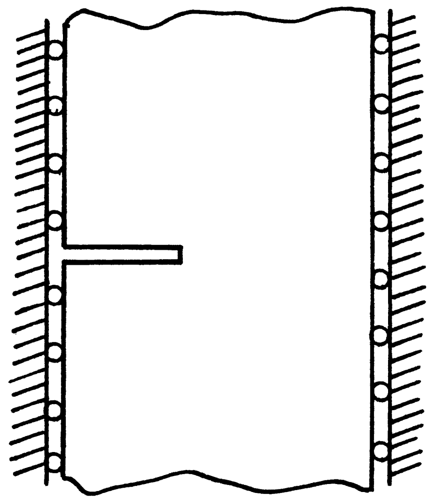


$$K_I = \frac{P(1-\alpha)}{\sqrt{W}} \sqrt{\tan\frac{\pi a}{W}} \frac{\cos\frac{\pi x_0}{W}}{\sqrt{\left(\sin\frac{\pi x_0}{W}\right)^2 - \left(\sin\frac{\pi a}{W}\right)^2}}$$

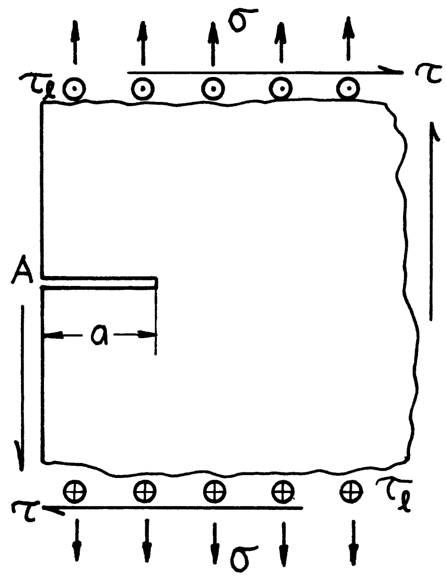
where

$$\alpha = \begin{cases} \frac{1}{2}(1+\nu) & \text{plane stress} \\ \frac{1}{2}\left(\frac{1}{1-\nu}\right) & \text{plane strain} \end{cases}$$

Method: Integration of **page 7.7**
Accuracy: Exact
Reference: **Tada 2000**



Equivalent to Periodic Cracks with Symmetric Loadings



Stress Intensity Factors

$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = 1.1215 \binom{\sigma}{\tau} \sqrt{\pi a}$$

$$K_{III} = \tau_\ell \sqrt{\pi a}$$

Displacements at A

$$\begin{Bmatrix} 2v \\ 2u \end{Bmatrix} = 1.454 \cdot \frac{4}{E'} \begin{Bmatrix} \sigma \\ \tau \end{Bmatrix} a = \frac{5.816}{E'} \begin{Bmatrix} \sigma \\ \tau \end{Bmatrix} a$$

$$2w = \frac{2\tau_\ell a}{G}$$

Methods and References: K_I Integral Transform-Singular Integral Equation (**Wigglesworth 1957; Koiter 1965a, Bueckner 1966; Sneddon 1971a; Benthem 1972**); Successive Stress Relaxation-Integral Equation (**Irwin 1958b, 1960a; Lachenbruch 1961; Nishitani 1971a; Tada 1972a**)

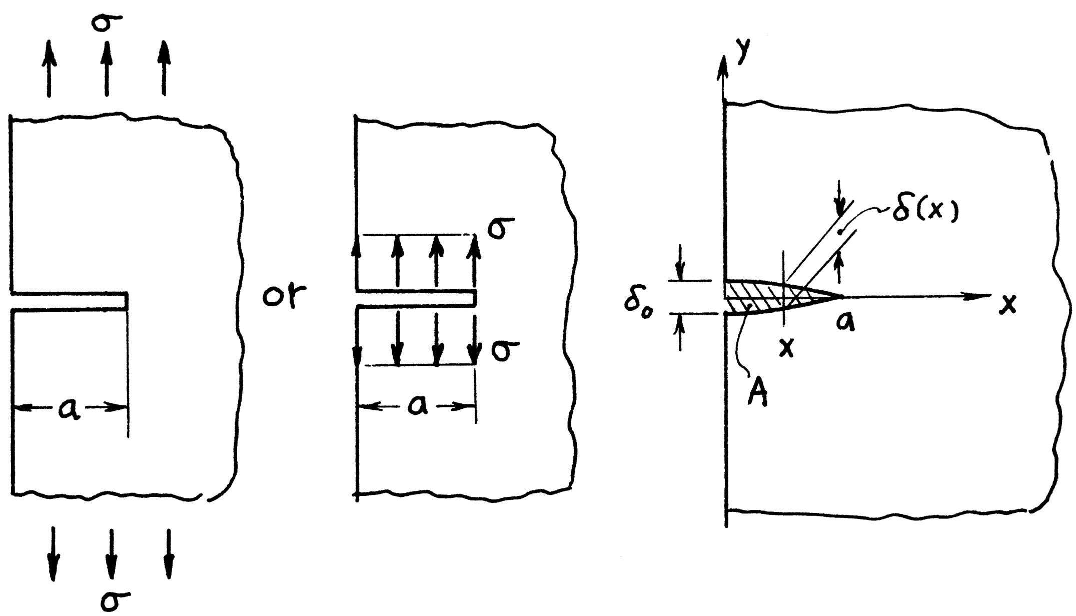
K_{II} From similarity of free boundary corrections between Mode I and Mode II (**Tada 1973**)

K_{III} Westergaard Stress Function, etc.

Accuracy: K_I, K_{II} Within one unit of last digit

K_{III} Exact

Displacements were derived by Paris' equation (**Paris 1957**) (see **Appendix B**)



$$K_I = 1.1215\ \sigma\sqrt{\pi a}$$

Crack Opening Area:

$$A = 1.258\ \frac{\sigma \pi a^2}{E'}$$

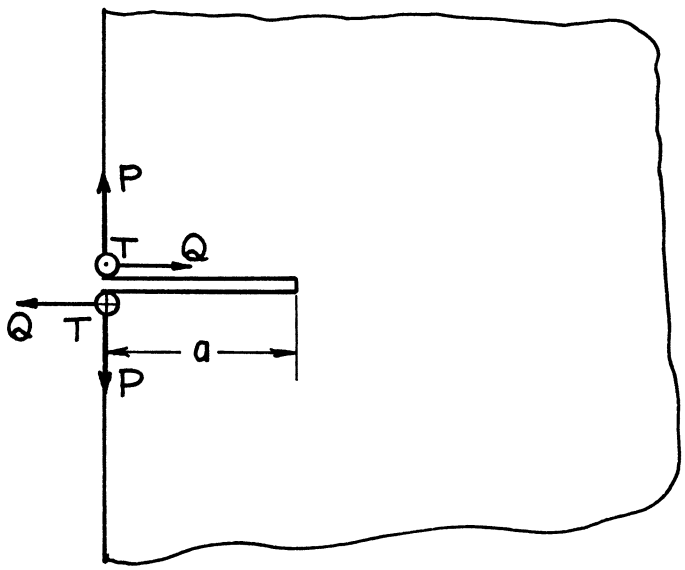
Crack Profile:

$$\delta(x) = \frac{4\sigma}{E'}\sqrt{a^2 - x^2}\ D\left(\frac{x}{a}\right)$$

$$D\left(\frac{x}{a}\right) = 1.454 - .727\frac{x}{a} + .618\left(\frac{x}{a}\right)^2 - .224\left(\frac{x}{a}\right)^3$$

$$\delta_0 = \delta(0) = 1.454\frac{4\sigma a}{E'}$$

Method: A and δ Paris' Equation (see **Appendix B**)
Accuracy: A and δ_0 Within one unit of last digit; $\delta(x)$ 1%
Reference: **Tada 1985**

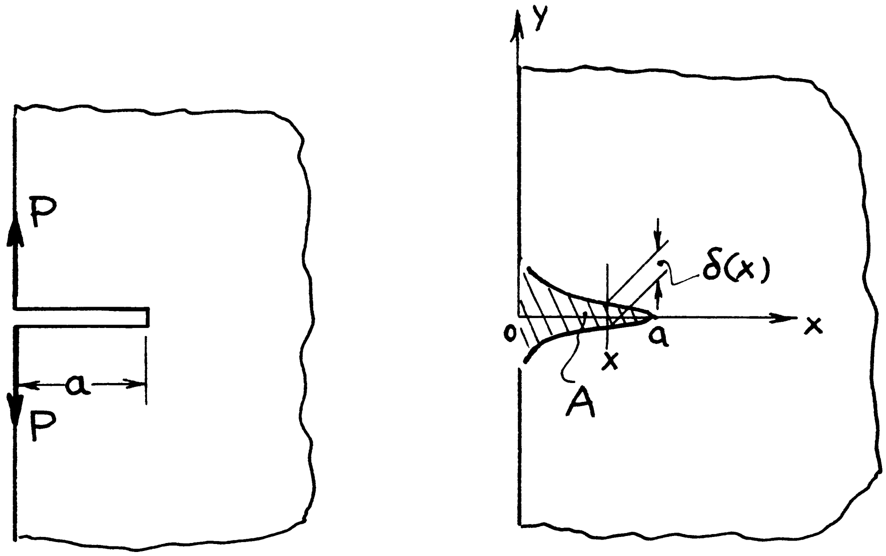


$$K_I = 1.297 \cdot \frac{2P}{\sqrt{\pi a}}$$

$$K_{II} = 1.297 \cdot \frac{2Q}{\sqrt{\pi a}} \qquad \left(\frac{\pi}{\sqrt{\pi^2 - 4}} = 1.297\right)$$

$$K_{III} = \frac{2T}{\sqrt{\pi a}}$$

Methods: K_I, K_{II} Combined Method of Integral Transform and Paris' Equation (see **Appendix B**);
K_{III} Westergaard Stress Function, etc.
Accuracy: Exact
References: **Ouchterlony 1975; Tada 1973, 1985**



$$K_I = 1.297 \frac{2P}{\sqrt{\pi a}}$$

Crack Opening Area:

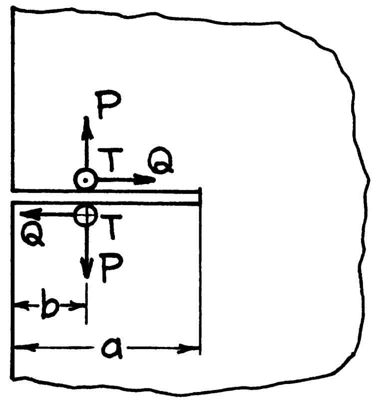
$$A = 1.454 \frac{4Pa}{E'}$$

Crack Profile:

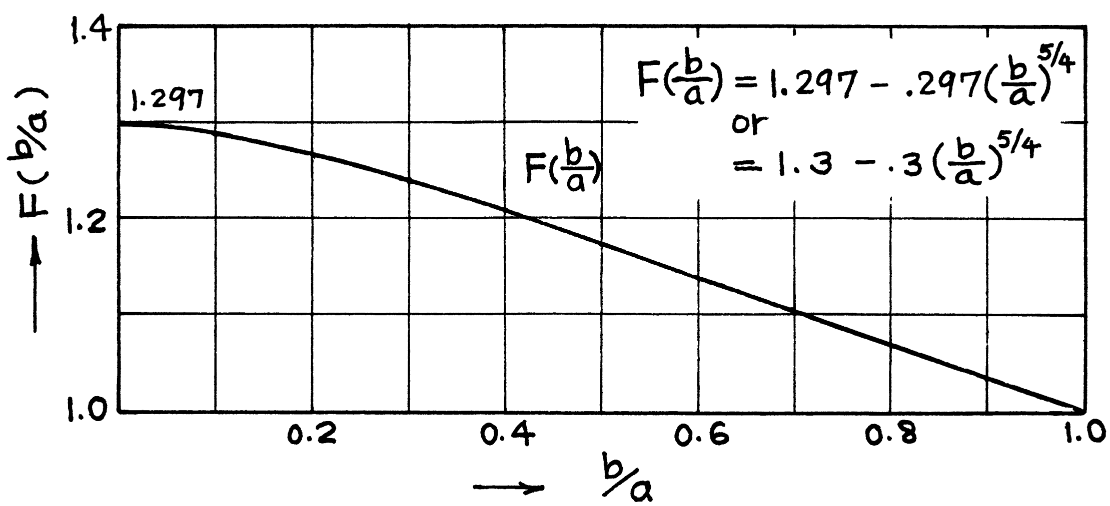
$$\delta(x) = \frac{8P}{\pi E'} \cosh^{-1} \frac{a}{x} \cdot D\left(\frac{x}{a}\right)$$

$$D\left(\frac{x}{a}\right) = 1.681 - .384\left(\frac{x}{a}\right)^{.38}$$

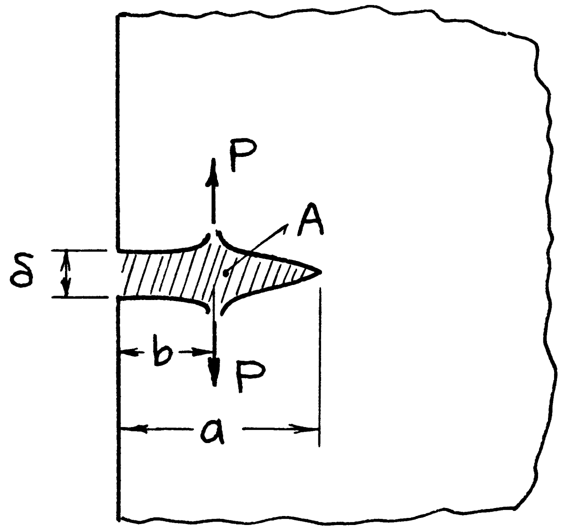
Method: A and δ Paris' Equation (see **Appendix B**)
Accuracy: A Within one unit of last digit; δ 1%
Reference: **Tada 1985**



$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \frac{2}{\sqrt{\pi a}} \begin{Bmatrix} P \\ Q \\ T \end{Bmatrix} \frac{1}{\sqrt{1-(b/a)^2}} \begin{Bmatrix} F(b/a) \\ F(b/a) \\ 1 \end{Bmatrix}$$



Methods: K_I, K_{II} Alternating Method (Successive Stress Adjustment)
K_{III} Westergaard Stress Function, etc.
Accuracy: K_I, K_{II} 0.5%
K_{III} Exact
References: **Hartranft 1973; Tada 1973, 1985**



$$K_I = \frac{2P}{\sqrt{\pi a}} F\left(\frac{b}{a}\right)$$

$$F\left(\frac{b}{a}\right) = \frac{1.3 - .3\left(\frac{b}{a}\right)^{5/4}}{\sqrt{1 - \left(\frac{b}{a}\right)^2}}$$

Crack Opening Area:

$$A = \frac{4P}{E'} \sqrt{a^2 - b^2} \cdot G\left(\frac{b}{a}\right)$$

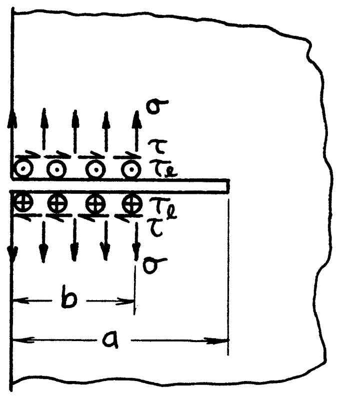
$$G\left(\frac{b}{a}\right) = 1.454 - .727\frac{b}{a} + .618\left(\frac{b}{a}\right)^2 - .224\left(\frac{b}{a}\right)^3$$

Opening at Edge:

$$\delta = \frac{8P}{\pi E'} \cosh^{-1}\frac{a}{b} \cdot H\left(\frac{b}{a}\right)$$

$$H\left(\frac{b}{a}\right) = 1.681 - .384\left(\frac{b}{a}\right)^{.38}$$

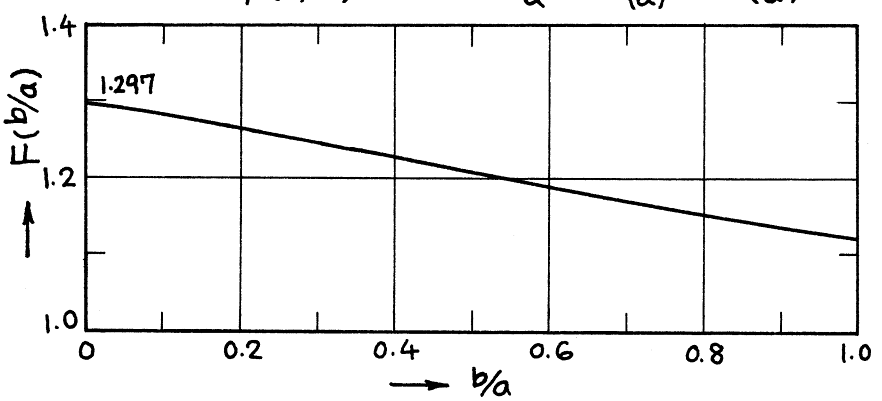
Method: A and δ Paris' Equation (see **Appendix B**)
Accuracy: A and δ 1%
Reference: **Tada 1985**



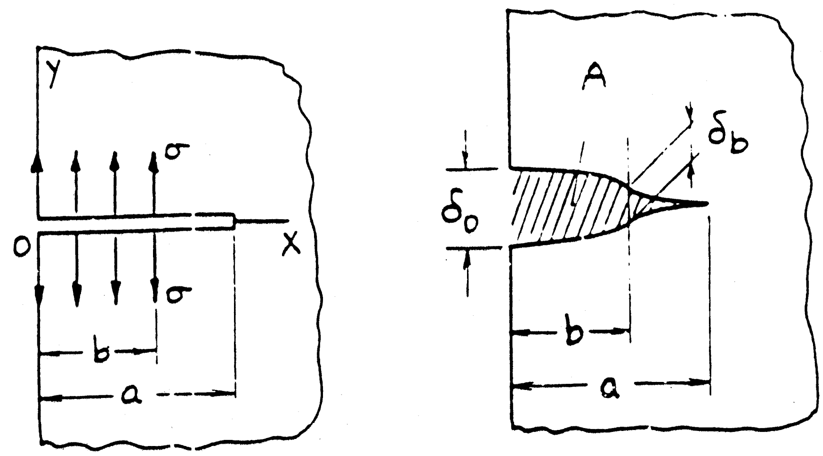
$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III}\end{Bmatrix} = \begin{Bmatrix}\sigma\\ \tau \\ \tau_\ell\end{Bmatrix}\sqrt{\pi a}\cdot\frac{2}{\pi}\sin^{-1}\frac{b}{a}\cdot\begin{Bmatrix}F(b/a)\\ F(b/a)\\ 1\end{Bmatrix}$$

$$F(b/a) = 1.3 - 0.18\frac{b}{a} \qquad (1)$$

$$\text{or } F(b/a) = 1.3 - .143\frac{b}{a} - .120\left(\frac{b}{a}\right)^2 + .083\left(\frac{b}{a}\right)^3 \qquad (2)$$



Methods: K_I, K_{II} Alternating Method or Integration of **page 8.3;**
K_{III} Westergaard Stress Function, etc.
Accuracy: Empirical Formula (1) 0.5%, (2) 0.2%; K_{III} Exact
References: **Hartranft 1973; Tada 1985**



$$K_I = \sigma\sqrt{\pi a}\cdot\frac{2}{\pi}\sin^{-1}\frac{b}{a}\cdot F\left(\frac{b}{a}\right)$$

$$F\left(\frac{b}{a}\right) = 1.3 - .18\frac{b}{a}$$

or

$$F\left(\frac{b}{a}\right) = 1.3 - .143\frac{b}{a} - .120\left(\frac{b}{a}\right)^2 + .083\left(\frac{b}{a}\right)^3$$

Crack Opening Area:

$$A = \frac{\sigma\pi a^2}{E'}\cdot\frac{2}{\pi}\left(\sin^{-1}\frac{b}{a} + \frac{b}{a}\sqrt{1-\left(\frac{b}{a}\right)^2}\right)\cdot G\left(\frac{b}{a}\right)$$

or

$$= \frac{\sigma\pi a^2}{E'}\cdot\overline{G}\left(\frac{b}{a}\right)$$

$$G\left(\frac{b}{a}\right) = 1.258 + .196\left(1-\frac{b}{a}\right)^{5/2}$$

$$\overline{G}\left(\frac{b}{a}\right) = \left(1.454 - .402\frac{b}{a}\right)\cdot\frac{2}{\pi}\sin^{-1}\frac{b}{a} + .926\,\frac{b}{a}\sqrt{1-\left(\frac{b}{a}\right)^2} + \left(\frac{b}{a}\right)^2\left(.206 - .257\cosh^{-1}\frac{a}{b}\right)$$

Opening at Edge:

$$\delta_0 = \frac{4\sigma a}{E'} \cdot \frac{2}{\pi}\left(\sin^{-1}\frac{b}{a} + \frac{b}{a}\cosh^{-1}\frac{a}{b}\right) \cdot H_0\left(\frac{b}{a}\right)$$

or

$$= \frac{4\sigma a}{E'} \cdot \overline{H}_0\left(\frac{b}{a}\right)$$

$$H_0\left(\frac{b}{a}\right) = 1.681 - .227\frac{b}{a}\left[1 + \left(1 - \frac{b}{a}\right)\left(\frac{b}{a}\right)^{-3/4}\right]$$

$$\overline{H}_0\left(\frac{b}{a}\right) = 1.681\frac{2}{\pi}\left(\sin^{-1}\frac{b}{a} + \frac{b}{a}\cosh^{-1}\frac{a}{b}\right) - .392\frac{b}{a} + .140\left(\frac{b}{a}\right)^2 + .025\left(\frac{b}{a}\right)^3$$

Opening at $x = b$:

$$\delta_b = \frac{4\sigma a}{E'} \cdot \frac{2}{\pi}\left[\sqrt{1 - \left(\frac{b}{a}\right)^2}\sin^{-1}\frac{b}{a} - \frac{b}{a}\ell n\frac{b}{a}\right] \cdot H_b\left(\frac{b}{a}\right)$$

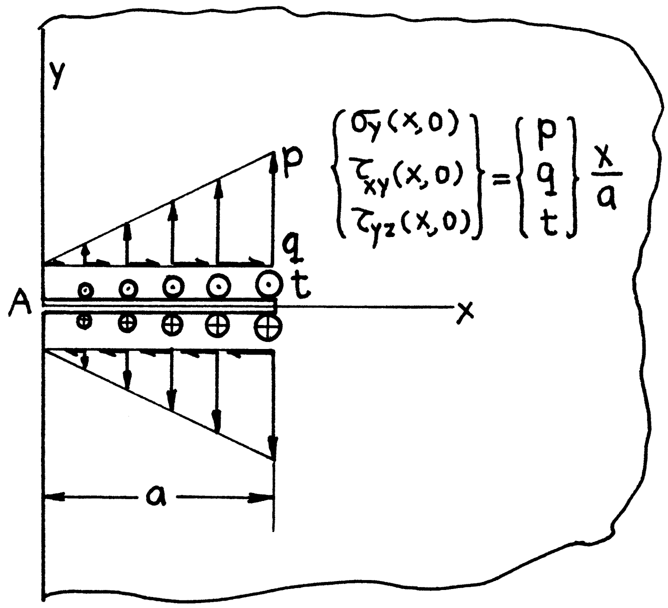
or

$$= \frac{4\sigma a}{E'} \cdot \overline{H}_b\left(\frac{b}{a}\right)$$

$$H_b\left(\frac{b}{a}\right) = 1.681\left[1 - .215\left(\frac{b}{a}\right)^{.320}\left(1 - \frac{b}{a}\right)^{.565}\right]$$

$$\overline{H}_b\left(\frac{b}{a}\right) = 1.681\frac{2}{\pi}\left[\sqrt{1 - \left(\frac{b}{a}\right)^2}\sin^{-1}\frac{b}{a} - \frac{b}{a}\ell n\frac{b}{a}\right]$$
$$- \frac{b}{a}\left(1 - \frac{b}{a}\right)\left[.382 + .143\left(\frac{b}{a}\right)^2\left(1 - \frac{b}{a}\right)\right]$$

Method: A and δ Paris' Equation (see **Appendix B**) or Integration of **page 8.3a**
Accuracy: G 2%, $\overline{G}$ 0.5%; H_0 1%, $\overline{H}_0$ 0.5%; H_b and $\overline{H}_b$ 1%
Reference: **Tada 1985**



Stress Intensity Factors

$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = 0.683 \begin{Bmatrix} p \\ q \end{Bmatrix} \sqrt{\pi a} = 1.072 \frac{2}{\pi} \begin{Bmatrix} p \\ q \end{Bmatrix} \sqrt{\pi a}$$

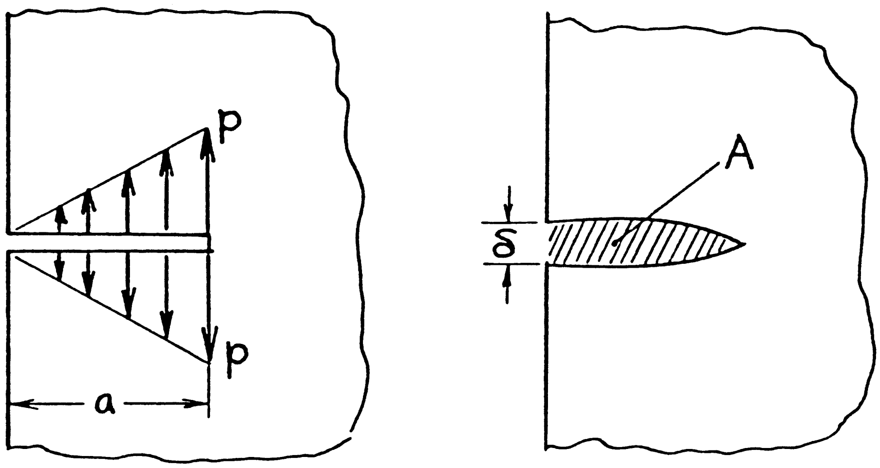
$$K_{III} = \frac{2}{\pi} t \sqrt{\pi a}$$

Displacements at A

$$\begin{Bmatrix} 2v \\ 2u \end{Bmatrix} = \frac{1.770}{E'} \begin{Bmatrix} p \\ q \end{Bmatrix} a = 1.390 \cdot \frac{4}{E'} \cdot \frac{1}{\pi} \begin{Bmatrix} p \\ q \end{Bmatrix} a$$

$$2w = \frac{2}{G} \cdot \frac{1}{\pi} ta$$

Method: Superposition of **page 8.1** and **page 8.6**
Accuracy: K_I, K_{II} 0.2%
K_{III} Exact
Reference: **Tada 1973**
Displacements were derived by Paris' equation (See **Appendix B**).



$$K_I = \left(\frac{2}{\pi} p\sqrt{\pi a}\right)(1.072) = .683 p\sqrt{\pi a}$$

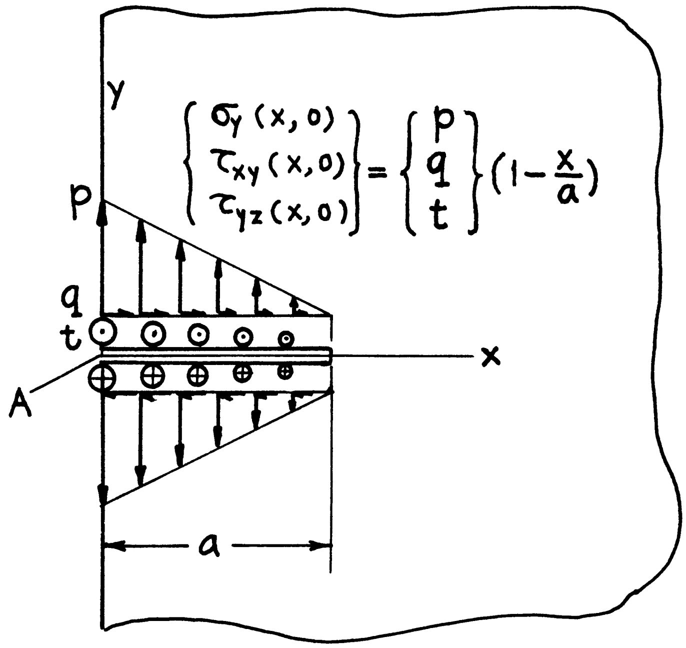
Crack Opening Area:

$$A = \left(\frac{4pa^2}{3E'}\right)(1.202) = 1.603\frac{pa^2}{E'}$$

Opening at Edge:

$$\delta = \left(\frac{4pa}{\pi E'}\right)(1.390) = 1.770\frac{pa}{E'}$$

Method: A and δ Paris' Equation (see **Appendix B**) or Integration of **page 8.3a**
Accuracy: Within one unit of last digit
Reference: **Tada 1985**



Stress Intensity Factors

$$\left\{\begin{matrix}K_I\\ K_{II}\end{matrix}\right\} = 0.439\left\{\begin{matrix}p\\ q\end{matrix}\right\}\sqrt{\pi a} = 1.208\left(1-\frac{2}{\pi}\right)\left\{\begin{matrix}p\\ q\end{matrix}\right\}\sqrt{\pi a}$$

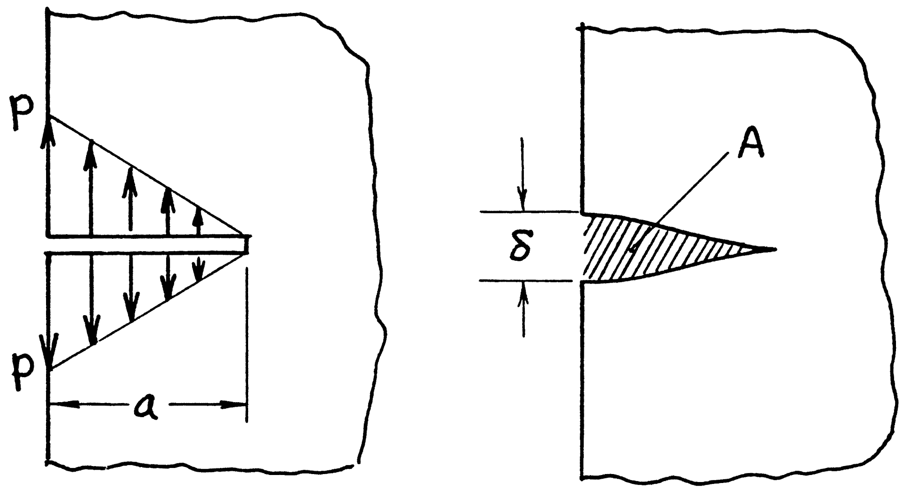
$$K_{III} = \left(1-\frac{2}{\pi}\right)t\sqrt{\pi a}$$

Displacements at A

$$\left\{\begin{matrix}2v\\ 2u\end{matrix}\right\} = \frac{4.046}{E'}\left\{\begin{matrix}p\\ q\end{matrix}\right\}a = 1.484\cdot\frac{4}{E'}\left(1-\frac{1}{\pi}\right)\left\{\begin{matrix}p\\ q\end{matrix}\right\}a$$

$$2w = \frac{2}{G}\left(1-\frac{1}{\pi}\right)ta$$

Methods: K_I Integral Transform (Benthem)
K_{II} From Similarity of Free Boundary Corrections (Tada)
K_{III} Westergaard Stress Function, etc., Stress Concentration Factor (Neuber)
Accuracy: K_I, K_{II} 0.2%
K_{III} Exact
References: **Neuber 1937; Benthem 1972; Tada 1973**
Displacements were derived by Superposition of **page 8.1** and **page 8.5.**



$$K_I = .439p\sqrt{\pi a} = \left(1 - \frac{2}{\pi}\right)p\sqrt{\pi a} \quad (1.208)$$

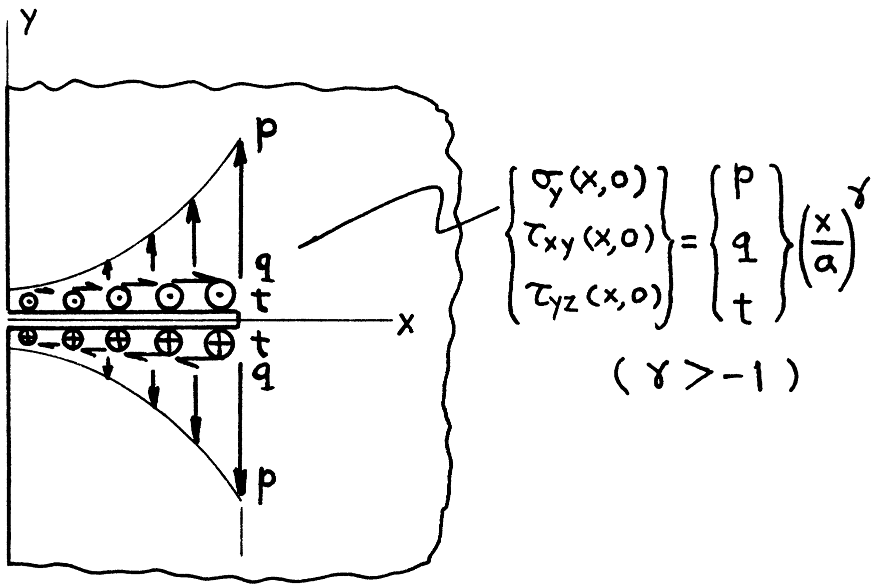
Crack Opening Area:

$$A = 2.349\,\frac{pa^2}{E'} = \left(\pi - \frac{4}{3}\right)\frac{pa^2}{E'} \quad (1.299)$$

Opening at Edge:

$$\delta = 4.046\,\frac{pa}{E'} = 4\left(1 - \frac{1}{\pi}\right)\frac{pa}{E'} \quad (1.484)$$

Method: A and δ Superposition of **page 8.1a** and **page 8.5a**
Accuracy: Within one unit of last digit
Reference: **Tada 1985**



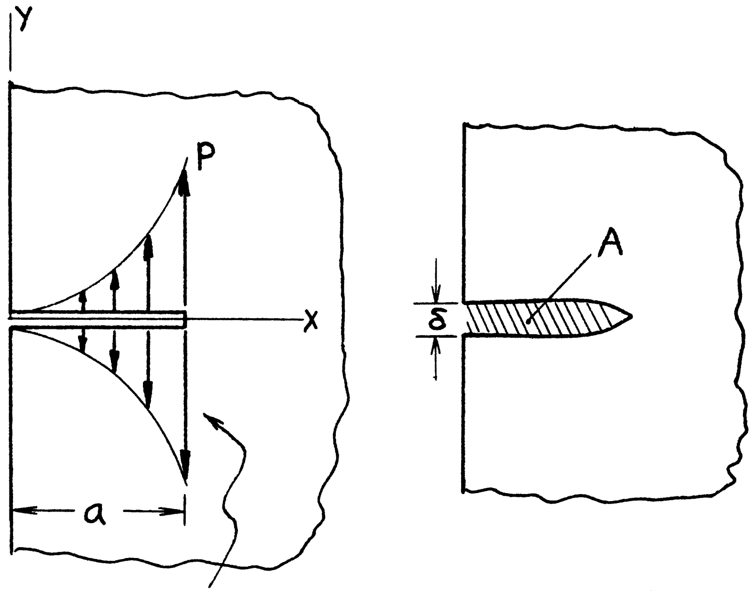
$$\begin{Bmatrix} K_I \\ K_{II} \\ K_{III} \end{Bmatrix} = \begin{Bmatrix} p \\ q \\ t \end{Bmatrix} \sqrt{a} \frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{\Gamma\left(\frac{\gamma}{2}+1\right)} \cdot \begin{Bmatrix} F(\gamma) \\ F(\gamma) \\ 1 \end{Bmatrix}$$

$$F(\gamma) = 1 + \frac{.188\gamma + .488}{(\gamma+2)^2}$$

$\Gamma(\gamma)$ = Gamma Function (See **Appendix M**)

Methods: Integral Equation (Stallybrass), Alternating Method (Hartranft), Integration of **page 8.3**
Accuracy: K_I and K_{II} 0.5%
K_{III} Exact
References: **Stallybrass 1970; Hartranft 1973; Tada 1973, 1985**

NOTE: For special cases $\gamma = 0$ and $\gamma = 1$, see **page 8.1** and **page 8.5**, respectively. γ does not have to be an integer; it can be any real value $\gamma > -1$.



$$\sigma_y(x,0) = p\left(\frac{x}{a}\right)^{\gamma} \ (\gamma > -1)$$

$$K_I = p\sqrt{a}\,\frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{\Gamma\left(\frac{\gamma}{2}+1\right)}\cdot F(\gamma)$$

Crack Opening Area:

$$A = \frac{pa^2}{E'}\sqrt{\pi}\,\frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{\Gamma\left(\frac{\gamma}{2}+2\right)}\cdot \{1.1215\cdot F(\gamma)\}$$

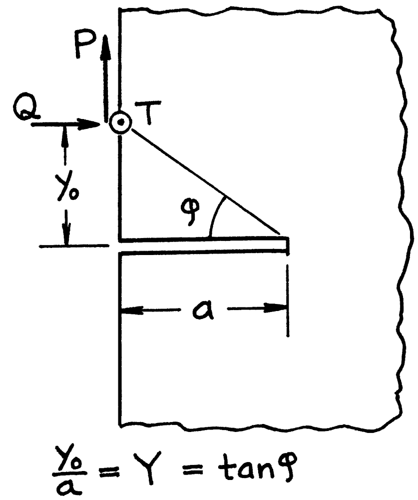
Opening at Edge:

$$\delta = \frac{4pa}{E'}\cdot\frac{1}{\sqrt{\pi}}\cdot\frac{\Gamma\left(\frac{\gamma+1}{2}\right)}{(\gamma+1)\Gamma\left(\frac{\gamma}{2}+1\right)}\cdot\{1.2967\cdot F(\gamma)\}$$

$$F(\gamma) = 1 + \frac{.188\gamma + .488}{(\gamma+2)^2}$$

$$\Gamma(\gamma) = \text{Gamma Function (See } \textbf{Appendix M}\text{)}$$

Method: A and δ Paris' Equation (see **Appendix B**) or Integration of **page 8.3a**
Accuracy: 0.5%
Reference: **Tada 1985**



$$\begin{Bmatrix} K_I \\ K_{II} \end{Bmatrix} = \frac{1}{\sqrt{\pi a}} \begin{Bmatrix} P \cdot F_1(Y) \\ Q \cdot F_3(Y) \end{Bmatrix} - \frac{1}{\sqrt{\pi a}} \cdot \frac{2}{\pi} \begin{Bmatrix} Q \\ P \end{Bmatrix} \cdot F_2(Y)$$

$$K_{III} = \frac{T}{\sqrt{\pi a}} \cdot \frac{1}{\sqrt{1+Y^2}} = \frac{T}{\sqrt{\pi a}} \cos\varphi$$

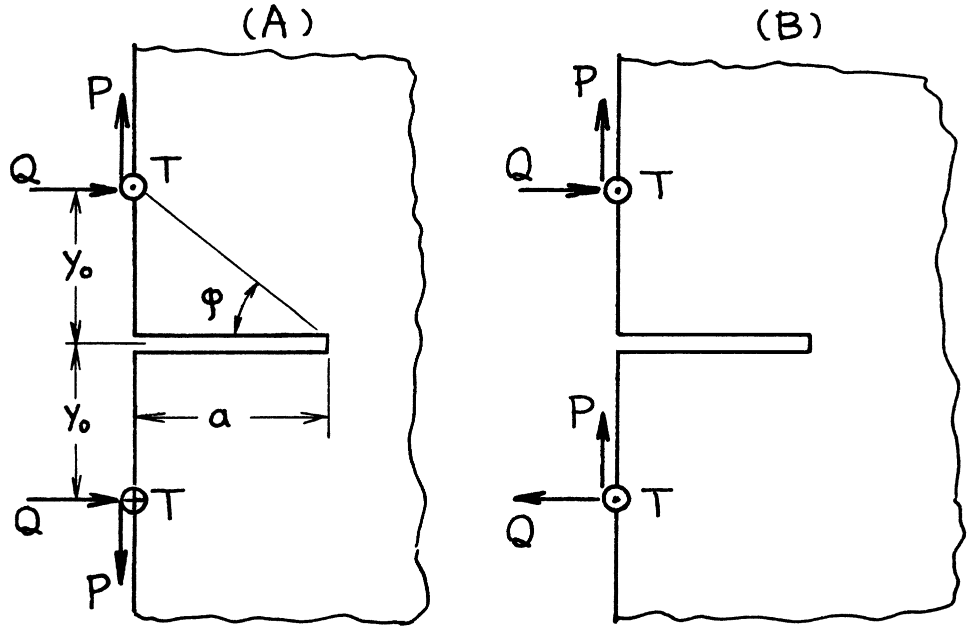
where

$$F_1(Y) = \frac{1+2Y^2}{\left(1+Y^2\right)^{3/2}} \left[1.3 - .3 \left(\frac{Y}{\sqrt{1+Y^2}} \right)^{5/4} \left\{ .665 - .267 \left(\frac{Y}{\sqrt{1+Y^2}} \right)^{5/4} \left(\frac{Y}{\sqrt{1+Y^2}} - .73 \right) \right\} \right]$$

$$F_2(Y) = \left[\frac{1}{1+Y^2} + \frac{Y^2}{\left(1+Y^2\right)^{3/2}} \tanh^{-1} \left(\frac{1}{\sqrt{1+Y^2}} \right) \right] \left[1.3 - .375 \frac{Y}{\sqrt{1+Y^2}} \left(1 - .4 \frac{Y}{\sqrt{1+Y^2}} \right) \right]$$

$$F_3(Y) = \frac{1}{\left(1+Y^2\right)^{3/2}} \left[1.3 - .75 \frac{Y}{\sqrt{1+Y^2}} \left(1 - 1.184 \frac{Y}{\sqrt{1+Y^2}} + .512 \frac{Y^2}{1+Y^2} \right) \right]$$

Method: Integration of **page 8.3**
Accuracy: Formulas $F_1(Y)$, $F_2(Y)$, and $F_3(Y)$ 0.5%
K_{III} Exact
References: **Tada 1985, 2000**



$$\frac{y_0}{a} = Y = \tan\varphi$$

$$(A) \qquad K_I = \frac{2}{\sqrt{\pi a}} P \cdot F_1(Y) - \frac{2}{\sqrt{\pi a}} \left\{ \frac{2}{\pi} Q \right\} \cdot F_2(Y)$$

$$K_{II} = 0$$

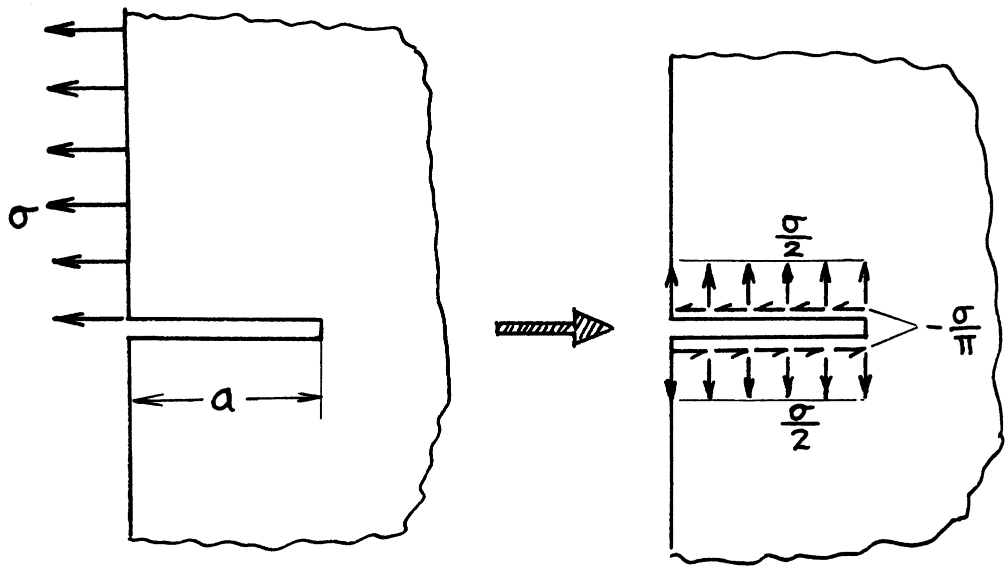
$$K_{III} = \frac{2}{\sqrt{\pi a}} T \cdot \frac{1}{\sqrt{1+Y^2}} = \frac{2}{\sqrt{\pi a}} T \cdot \cos\varphi$$

$$(B) \qquad K_I = 0$$

$$K_{II} = \frac{2}{\sqrt{\pi a}} Q \cdot F_3(Y) - \frac{2}{\sqrt{\pi a}} \left\{ \frac{2}{\pi} P \right\} \cdot F_2(Y)$$

$$K_{III} = 0$$

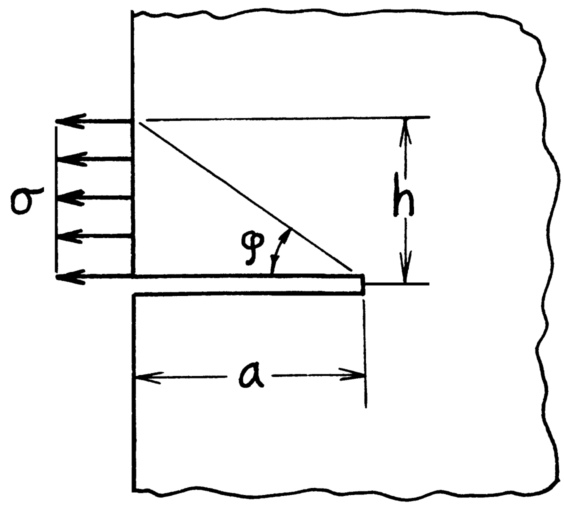
For $F_1(Y)$, $F_2(Y)$, and $F_3(Y)$, see **page 8.8**
Method: Superposition of **page 8.8**
Accuracy: Formulas $F_1(Y)$, $F_2(Y)$, and $F_3(Y)$ 0.5%
K_{II} and K_{III} in **(A)** and K_I and K_{III} in **(B)** are exact.
Reference: **Tada 1985, 2000**



$$K_I = \left(\frac{\sigma}{2}\right)\sqrt{\pi a}(1.1215) = .561\sigma\sqrt{\pi a}$$

$$K_{II} = \left(-\frac{\sigma}{\pi}\right)\sqrt{\pi a}(1.1215) = -.367\sigma\sqrt{\pi a}$$

Method: Direct use of **page 8.1** or Integration of **page 8.8**
Accuracy: The value 1.1215 is accurate within one unit of last digit.
References: **Tada 1985**



$$s = \frac{\varphi}{(\pi/2)} = \frac{2}{\pi}\tan^{-1}\frac{h}{a}$$

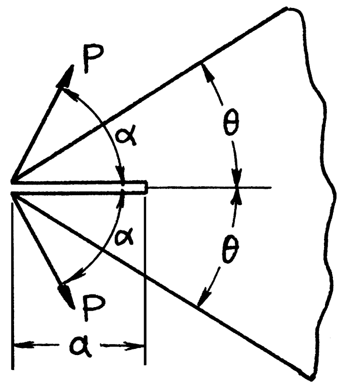
$$K_I = \left(\frac{1}{2}\sigma\right)\sqrt{\pi a}\cdot F_I(s)$$

$$K_{II} = \left(-\frac{1}{\pi}\sigma\right)\sqrt{\pi a}\cdot F_{II}(s)$$

$$F_I(s) = s\left[1.122 - (1-s)\left\{.296 + .25s^{3/4}(.75 - s)\right\}\right]$$

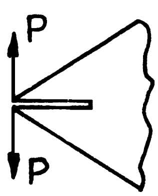
$$F_{II}(s) = s\left[1.122 + (1-s)\left(.915 + .26s^{3/2}\right)\right]$$

Method: Integration of **page 8.3** or **page 8.8**
Accuracy: 0.5%
References: **Tada 1985**



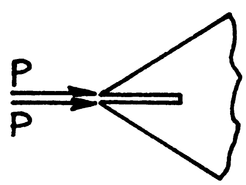
$$K_I = \frac{P}{\sqrt{a}}\left(\sin\alpha - \cos\alpha \frac{2\sin^2\theta}{2\theta + \sin 2\theta}\right)\sqrt{\frac{2\theta + \sin 2\theta}{\theta^2 - \sin^2\theta}}$$

$$K_I \gtreqless 0 : \alpha \gtreqless \tan^{-1}\left(\frac{2\sin^2\theta}{2\theta + \sin 2\theta}\right)$$



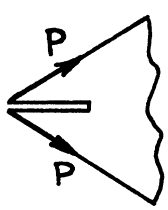
$$\alpha = \frac{\pi}{2}$$

$$K_I = \frac{P}{\sqrt{a}} \cdot \sqrt{\frac{2\theta + \sin 2\theta}{\theta^2 - \sin^2\theta}}$$



$$\alpha = 0$$

$$K_I = \frac{P}{\sqrt{a}} \cdot \frac{-2\sin^2\theta}{\sqrt{(2\theta + \sin 2\theta)\left(\theta^2 - \sin^2\theta\right)}}$$

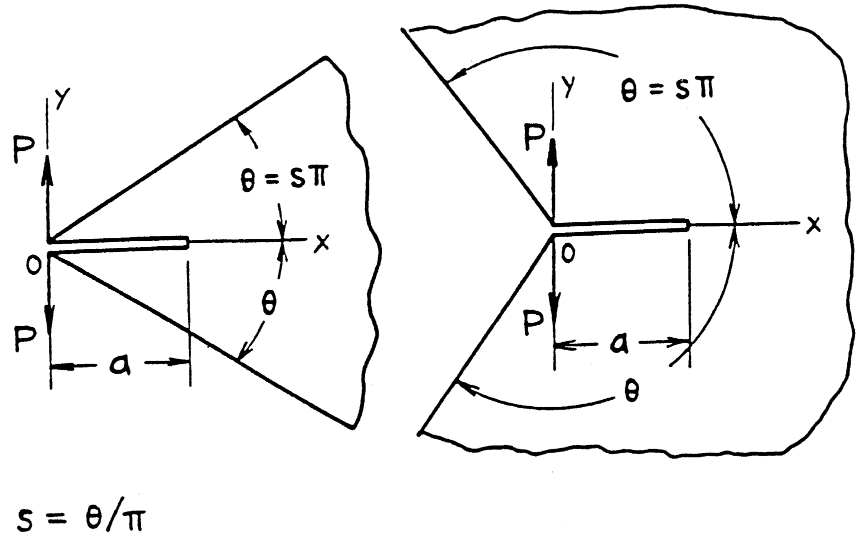


$$\alpha = \theta$$

$$K_I = \frac{P}{\sqrt{a}} \cdot \frac{2\theta \cdot \sin\theta}{\sqrt{(2\theta + \sin 2\theta)\left(\theta^2 - \sin^2\theta\right)}}$$

Method: Integral Transform
Accuracy: Exact
Reference: **Ouchterlony 1975**

NOTE: For special cases of $s = 1/2$ and $s = 1\left(\alpha = \pi/2\right)$, see **pages 8.2** and **3.6**, respectively.



$$K_I = \frac{P}{\sqrt{a}} F(s)$$

Crack Opening Area:

$$A = \frac{2Pa}{E'} \sqrt{\pi} F(s) \cdot G(s)$$

Crack Profile:

$$v(x, 0) = \frac{4P}{\pi E'} \sqrt{\pi} F(s) \cdot H\left(\frac{x}{a}, s\right)$$

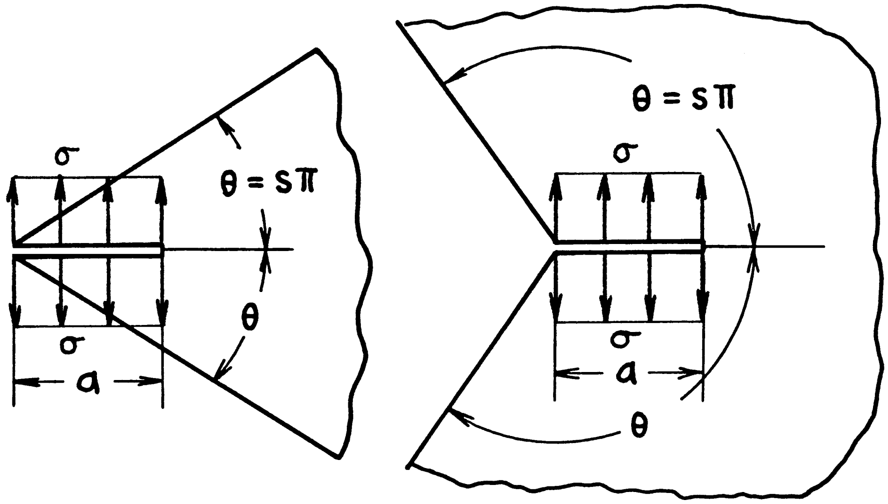
where

$$F(s) = \sqrt{\frac{2\pi s + \sin 2\pi s}{(\pi s)^2 - (\sin \pi s)^2}} = \sqrt{\frac{2\theta + \sin 2\theta}{\theta^2 - (\sin \theta)^2}}$$

$$G(s) = \frac{.1755 + .219s + .385s^2 + .120s^3}{s^{3/2}}$$

$$H\left(\frac{x}{a}, s\right) = \frac{1}{\sqrt{2}} \left[f(s) \tanh^{-1} \sqrt{1 - \frac{x}{a}} + g(s) \cdot \sqrt{1 - \frac{x}{a}} + h(s) \cdot \frac{1}{3}\left(2 + \frac{x}{a}\right) \sqrt{1 - \frac{x}{a}} \right]$$

For $f(s)$, $g(s)$ and $h(s)$, see **page 8.17**
Methods: K_I Combined method of Integral Transform and **Appendix B**
A and v Paris' Equation (see **Appendix B**) and Interpolation
Accuracy: K_I Exact; A and v 1%
References: **Ouchterlony 1975; Tada 1985**



$s = \theta/\pi$

$$K_I = \sigma\sqrt{\pi a} \cdot F(s)$$

Crack Opening Area:

$$A = \frac{\sigma \pi a^2}{E'} \{F(s)\}^2$$

Opening at Edge:

$$\delta = \frac{4\sigma a}{E'} F(s) \cdot G(s)$$

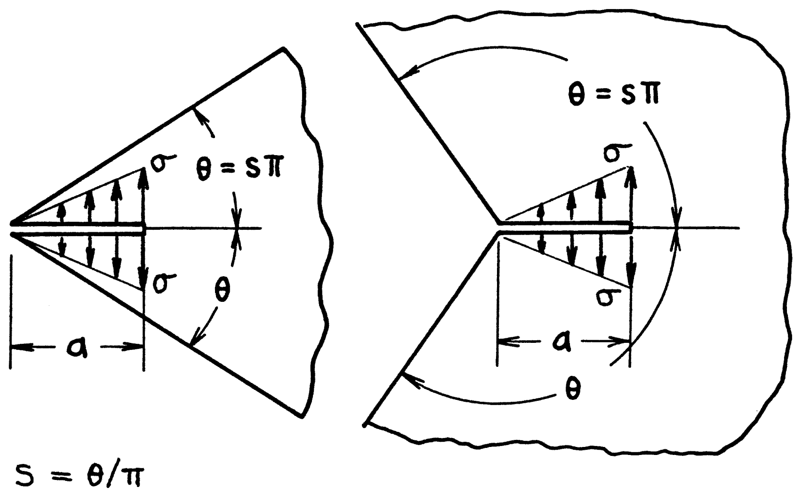
where

$$F(s) = \frac{\cdot 1755 + \cdot 219 s + \cdot 385 s^2 + \cdot 120 s^3}{s^{3/2}}$$

$$G(s) = \frac{\sqrt{\pi}}{2} \sqrt{\frac{2\pi s + \sin 2\pi s}{(\pi s)^2 - (\sin \pi s)^2}}$$

Methods: K_I Integral Transform and Integral Equation (**Doran** $s \geq \cdot 2$), Beam Theory ($s \to 0$), and Interpolation ($0 < s < .2$)
A and δ Paris' Equation (see **Appendix B**)
Accuracy: K_I and δ Better than 1%; A Better than 2%
References: **Doran 1969; Tada 1973, 1985**

NOTE: For special cases $s = 1/2$ and $s = 1$, see also **page 8.1** and **page 3.7**, respectively.



$$K_I = \sigma\sqrt{\pi a} \cdot F(s)$$

Crack Opening Area:

$$A = \frac{\sigma \pi a^2}{E'} \cdot \frac{2}{3} F(s) \cdot G(s)$$

Opening at Edge:

$$\delta = \frac{4\sigma a}{E'} \cdot F(s) \cdot H(s)$$

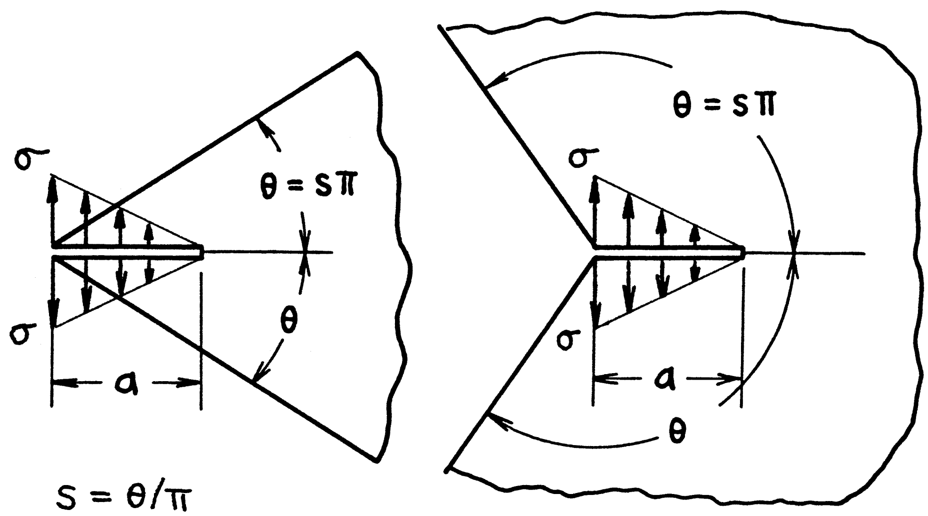
where

$$F(s) = \frac{.0585 + .196s + .333s^2 + .013s^3}{s^{3/2}}$$

$$G(s) = \frac{.1755 + .219s + .385s^2 + .120s^3}{s^{3/2}}$$

$$H(s) = \frac{\sqrt{\pi}}{4} \sqrt{\frac{2\pi s + \sin 2\pi s}{(\pi s)^2 - (\sin \pi s)^2}}$$

Methods: K_I Asymptotic Interpolation; A and δ Paris' Equation (see **Appendix B**)
Accuracy: K_I and δ Better than 1%; A Better than 2%
Reference: **Tada 1985**



$$K_I = \sigma\sqrt{\pi a} \cdot F(s)$$

Crack Opening Area:

$$A = \frac{\sigma \pi a^2}{E'} \cdot G(s) \cdot H(s)$$

Opening at Edge:

$$\delta = \frac{4\sigma a}{E'} \cdot I(s) \cdot J(s)$$

where

$$F(s) = \frac{.117 + .023s + .052s^2 + .107s^3}{s^{3/2}}$$

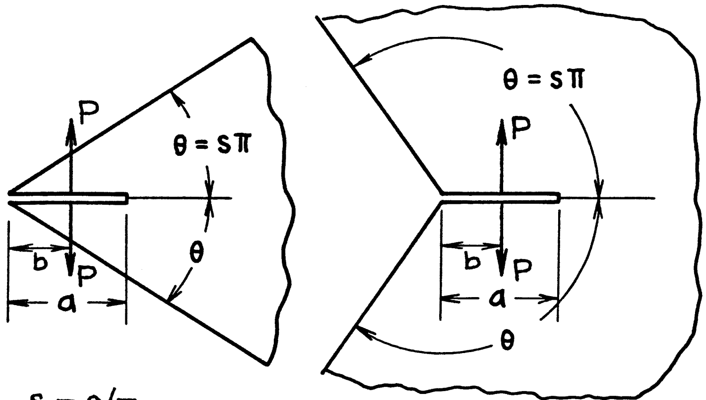
$$G(s) = \frac{.1755 + .219s + .385s^2 + .120s^3}{s^{3/2}}$$

$$H(s) = \frac{.1365 + .088s + .163s^2 + .111s^3}{s^{3/2}}$$

$$I(s) = \frac{.1463 + .121s + .219s^2 + .114s^3}{s^{3/2}}$$

$$J(s) = \frac{\sqrt{\pi}}{2}\sqrt{\frac{2\pi s + \sin 2\pi s}{(\pi s)^2 - (\sin \pi s)^2}}$$

Method: Superposition of **page 8.14** and **page 8.15**
Accuracy: K_I and δ Better than 1%; A Better than 2%
Reference: **Tada 1985**



s = θ/π

$$K_I = \frac{P}{\sqrt{a}} \cdot \sqrt{\frac{2}{\pi}} F\left(\frac{b}{a}, s\right)$$

Crack Opening Area:

$$A = \frac{4Pa}{E'} \cdot F_1(s) \cdot G\left(\frac{b}{a}, s\right)$$

Opening at Edge:

$$\delta = \frac{4P}{\pi E'} \sqrt{2} F_1(s) \cdot H\left(\frac{b}{a}, s\right)$$

where

$$F_1(s) = \frac{.1755 + .219s + .385s^2 + .120s^3}{s^{3/2}}$$

$$F\left(\frac{b}{a}, s\right) = \frac{f(s) + g(s)\frac{b}{a} + h(s)\left(\frac{b}{a}\right)^2}{\sqrt{1 - \frac{b}{a}}}$$

$$G\left(\frac{b}{a}, s\right) = \frac{1}{\sqrt{2}}\left[f(s)\left(\sqrt{1-\frac{b}{a}} + \frac{b}{a}\tanh^{-1}\sqrt{1-\frac{b}{a}}\right) + g(s)\cdot 2\frac{b}{a}\tanh^{-1}\sqrt{1-\frac{b}{a}} + h(s)\cdot 2\frac{b}{a}\sqrt{1-\frac{b}{a}}\right]$$

$$H\left(\frac{b}{a}, s\right) = f(s)\tanh^{-1}\sqrt{1-\frac{b}{a}} + g(s)\sqrt{1-\frac{b}{a}} + h(s)\cdot\frac{1}{3}\left(2+\frac{b}{a}\right)\sqrt{1-\frac{b}{a}}$$

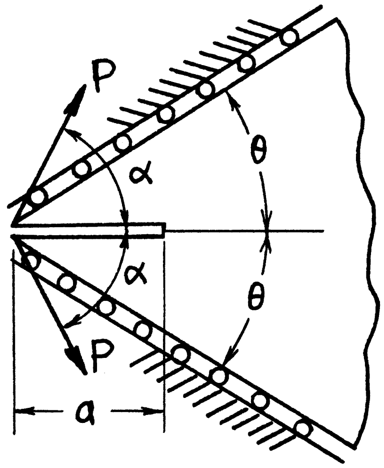
$$f(s) = \sqrt{\frac{\pi}{2}}\sqrt{\frac{2\pi s + \sin 2\pi s}{(\pi s)^2 - (\sin \pi s)^2}}$$

$$g(s) = -1 - 3f(s) + f_1(s)$$

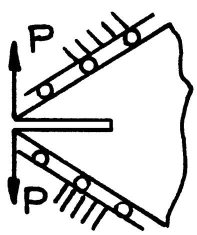
$$h(s) = 2 + 2f(s) - f_1(s)$$

$$f_1(s) = \left(1.103 + 3.615s^2 - .718s^3\right)/s^{3/2}$$

Method: Estimated by Interpolation
Accuracy: K_I and δ 2%; A 3%
Reference: **Tada 1985**

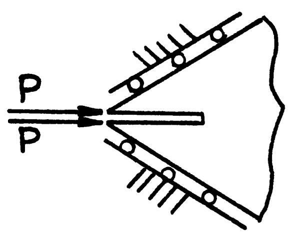


$$K_I = \frac{2P}{\sqrt{a}} \cdot \frac{\cos(\theta - \alpha)}{\sqrt{2\theta + \sin 2\theta}}$$



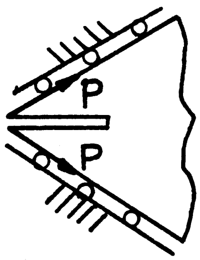
$$\alpha = \frac{\pi}{2}$$

$$K_I = \frac{2P}{\sqrt{a}} \cdot \frac{\sin\theta}{\sqrt{2\theta + \sin 2\theta}}$$



$$\alpha = 0$$

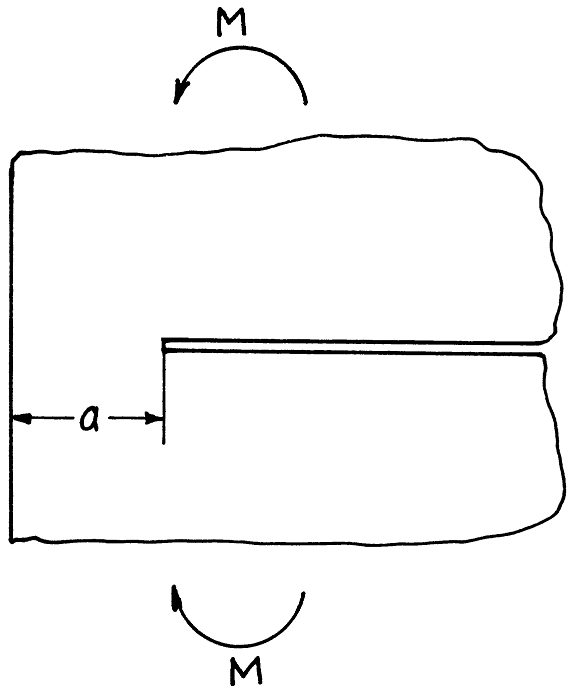
$$K_I = \frac{2P}{\sqrt{a}} \cdot \frac{\cos\theta}{\sqrt{2\theta + \sin 2\theta}}$$



$$\alpha = \theta$$

$$K_I = \frac{2P}{\sqrt{a}} \cdot \frac{1}{\sqrt{2\theta + \sin 2\theta}}$$

Method: Integral Transform
Accuracy: Exact
Reference: **Ouchterlony 1975**



Stress Intensity Factors

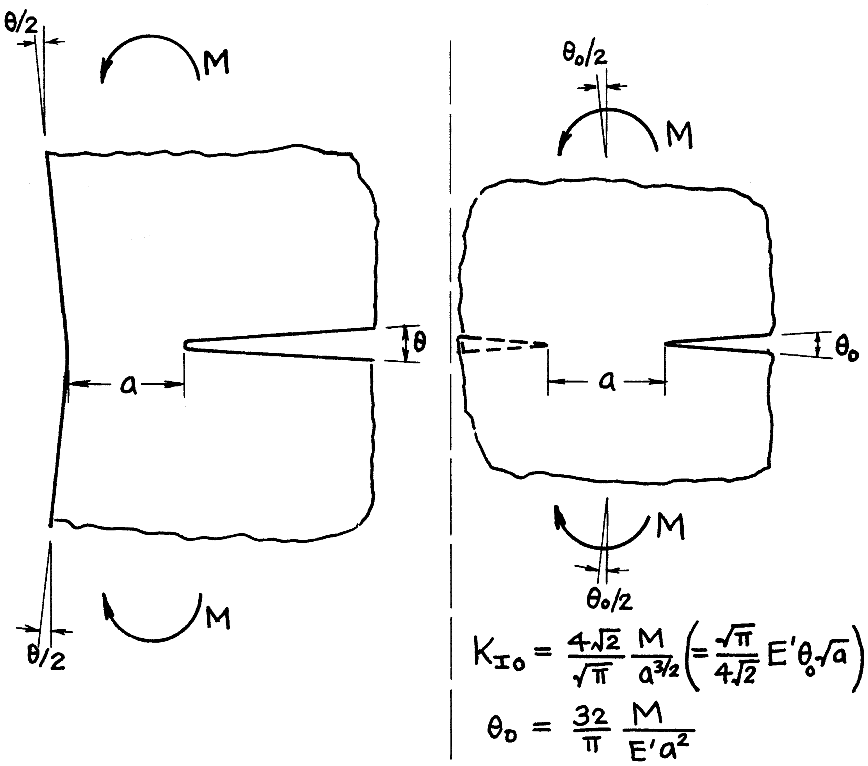
$$K_I = 3.975 \frac{M}{a\sqrt{a}} = \frac{E'\theta}{3.975} \cdot \sqrt{a}$$

$$K_{II} = K_{III} = 0$$

Displacement (Relative Rotation at Infinity)

$$\theta = \frac{15.8}{E'} \cdot \frac{M}{a^2}$$

Methods: Integral Transform (Benthem), Extrapolation from the Results by Boundary Collocation Method (Wilson)
Accuracy: Within 0.1%
References: **Wilson 1969, 1970; Benthem 1972**



$$K_I = 3.975\frac{M}{a^{3/2}} = 1.245K_{I0}$$

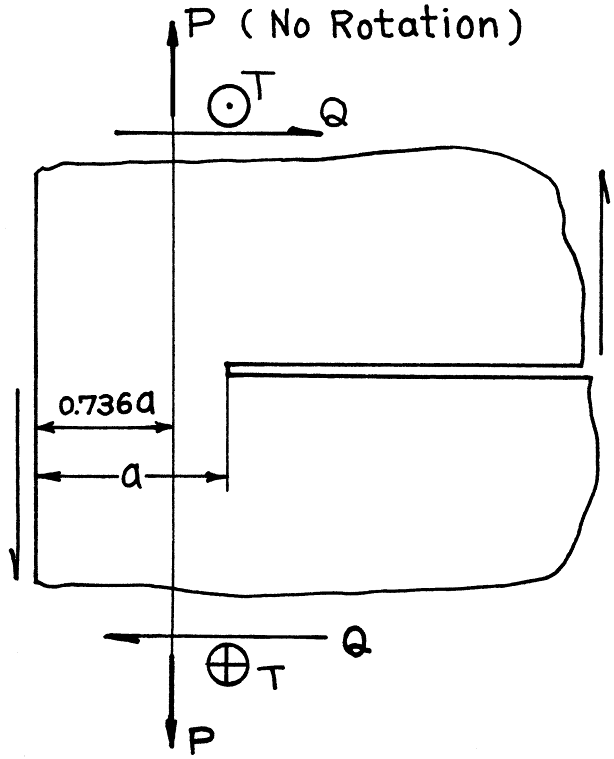
Relative Rotation at Infinity:

$$\theta = 15.80\frac{M}{E'a^2} = 1.551\theta_0\left(= 1.245^2\theta_0\right)$$

When θ is prescribed:

$$K_I = \frac{1}{3.975}E'\theta\sqrt{a} = \frac{1}{1.245}\left(\frac{\sqrt{\pi}}{4\sqrt{2}}E'\theta a\right) = \frac{K_{I0}}{1.245}$$

Method: (Comparison of **page 9.1** and **pages 4.10, 4.10a**)
Accuracy: Within one unit of last digit
Reference: **Tada 1985**

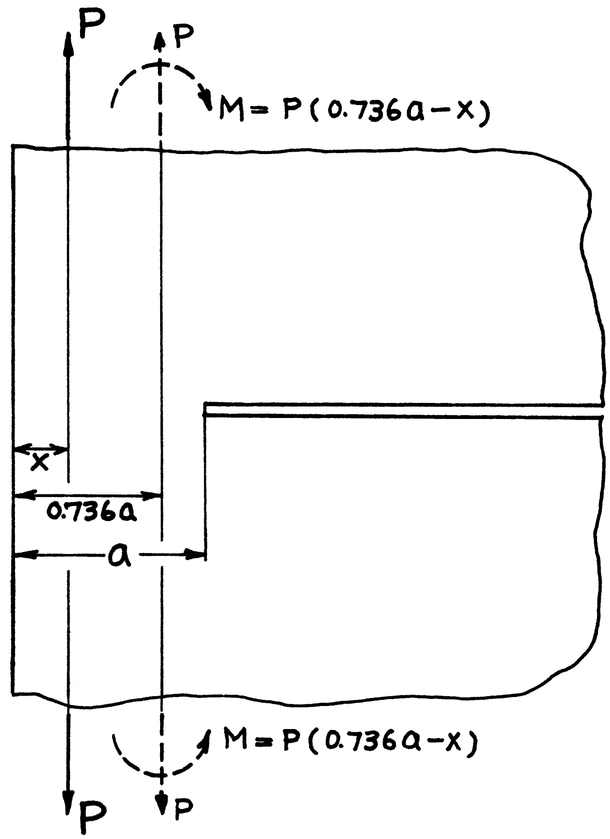


$$K_I = 1.297 \cdot \frac{2P}{\sqrt{\pi a}} \qquad \left(\frac{\pi}{\sqrt{\pi^2 - 4}} = 1.297 \right)$$

$$K_{II} = 1.297 \cdot \frac{2Q}{\sqrt{\pi a}}$$

$$K_{III} = \frac{2T}{\sqrt{\pi a}}$$

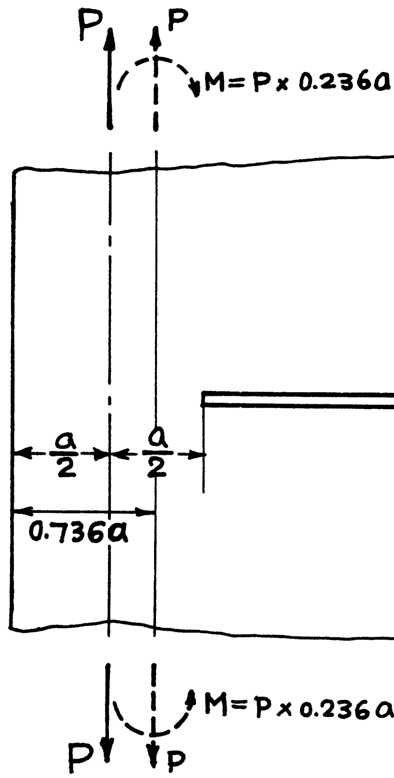
Methods: K_I Integral Transform (Benthem, Stallybrass), K_{II} Integral Equation (Tada), Boundary Collocation Method (Wilson), K_{III} Westergaard Stress Function, etc.
Accuracy: K_I 0.1%; K_{II}, K_{III} Exact
References: **Koiter 1965b; Wilson 1970; Stallybrass 1971; Benthem 1972; Tada 1973**



$$K_I = 3.522\left(\frac{x}{a} - 0.368\right)\frac{2P}{\sqrt{\pi a}}$$

$$K_{II} = K_{III} = 0$$

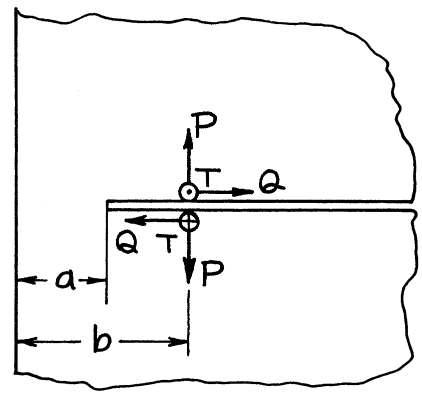
Method: Superposition of **page 9.1** and **page 9.2**
Accuracy: 0.1%
References: **Benthem 1972; Tada 1973**



$$K_I = 0.464 \cdot \frac{2P}{\sqrt{\pi a}}$$

$$K_{II} = K_{III} = 0$$

Method: Superposition of **page 9.1** and **page 9.2**
Accuracy: 0.2%
References: **Benthem 1972; Tada 1973**



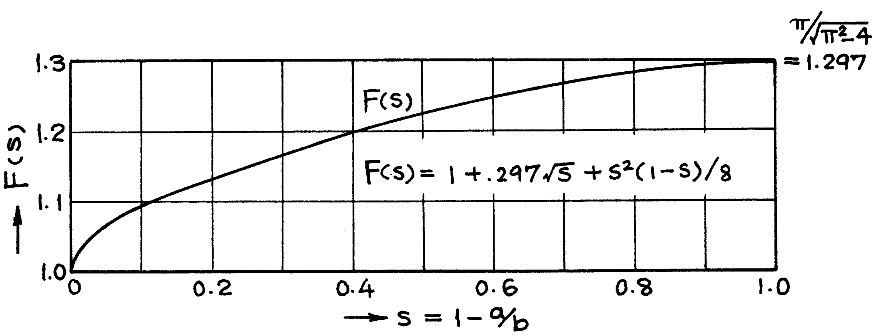
$$s = 1 - a/b$$

$$K_I = \frac{2P}{\sqrt{\pi a}} \cdot \frac{1}{\sqrt{1-\left(a/b\right)^2}} \cdot F(s) + 3.975 \cdot \frac{P(b-0.736a)}{a\sqrt{a}}$$

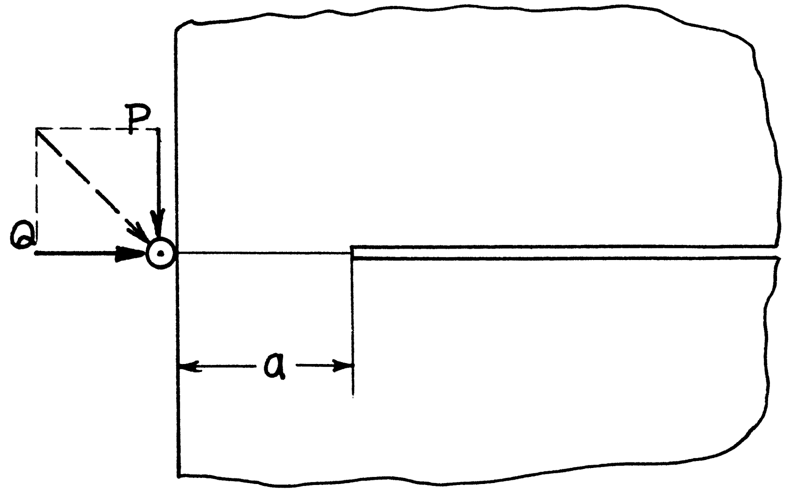
$$= \frac{2P}{\sqrt{\pi a}} \left\{ \frac{1}{\sqrt{1-\left(a/b\right)^2}} \cdot F(s) + 3.52\left(\frac{b}{a} - 0.736\right) \right\}$$

$$K_{II} = \frac{2Q}{\sqrt{\pi a}} \cdot \frac{1}{\sqrt{1-\left(a/b\right)^2}} \cdot F(s)$$

$$K_{III} = \frac{2T}{\sqrt{\pi a}} \cdot \frac{1}{\sqrt{1-\left(a/b\right)^2}}$$

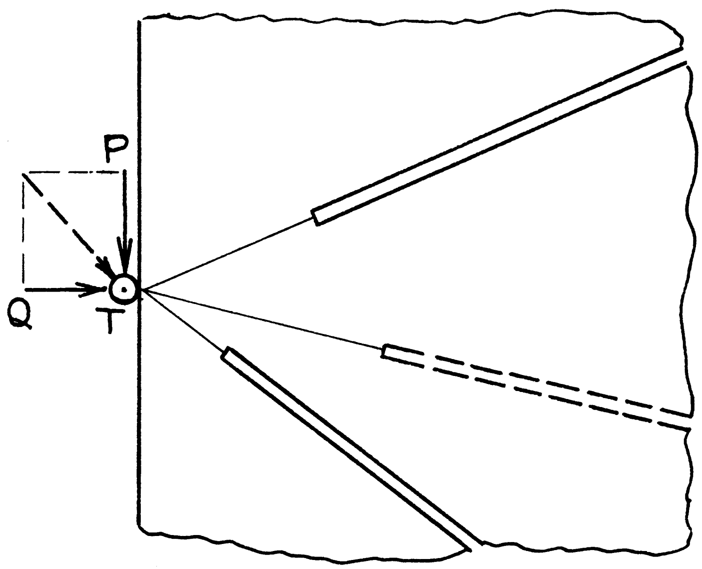


Method: K_I, K_{II} Successive Stress Relaxation - Integral Equations; K_{III} Westergaard Stress Function, etc.
Accuracy: K_I, K_{II} Estimated at 1%; K_{III} Exact
References: **Tada 1972a, 1973**



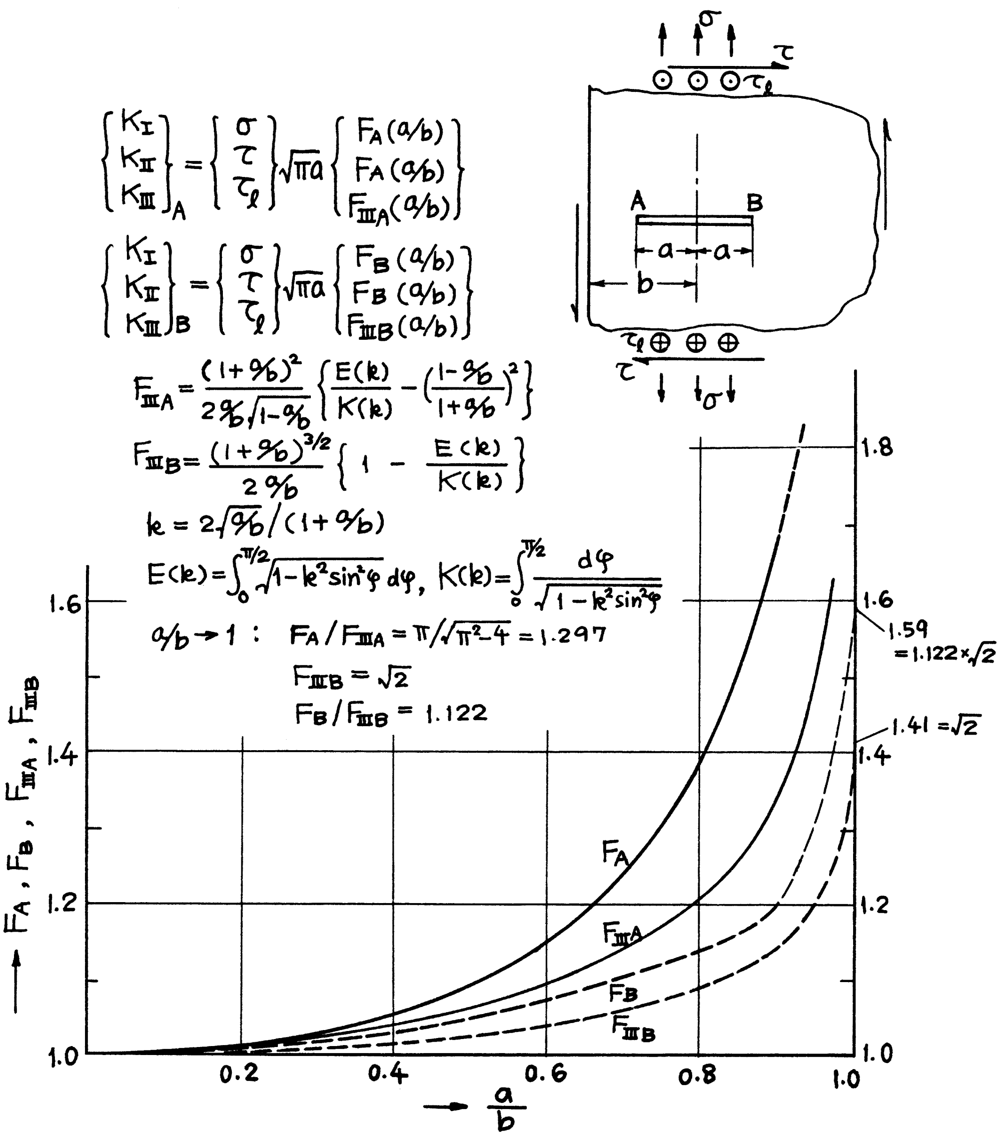
$$K_I = K_{II} = K_{III} = 0$$

Method: Special case of **page 9.6a**
Accuracy: Exact
Reference: **Tada 1973**



$$K_I = K_{II} = K_{III} = 0$$

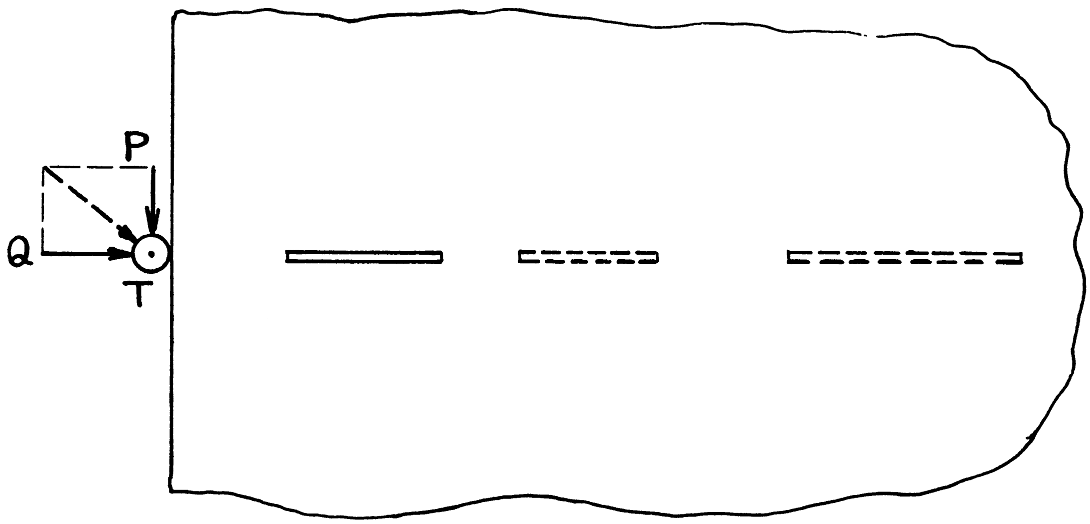
Methods: Westergaard Stress Function, etc. (Simple Radial Stresses for P and Q)
Accuracy: Exact
Reference: **Tada 1985**



Methods: K_I Expansions of Complex Stress Potentials (Isida)
K_{II} From similarity of free boundary corrections (Tada)
K_{III} Muskhelishvili Method, etc. (**page 6.1**)

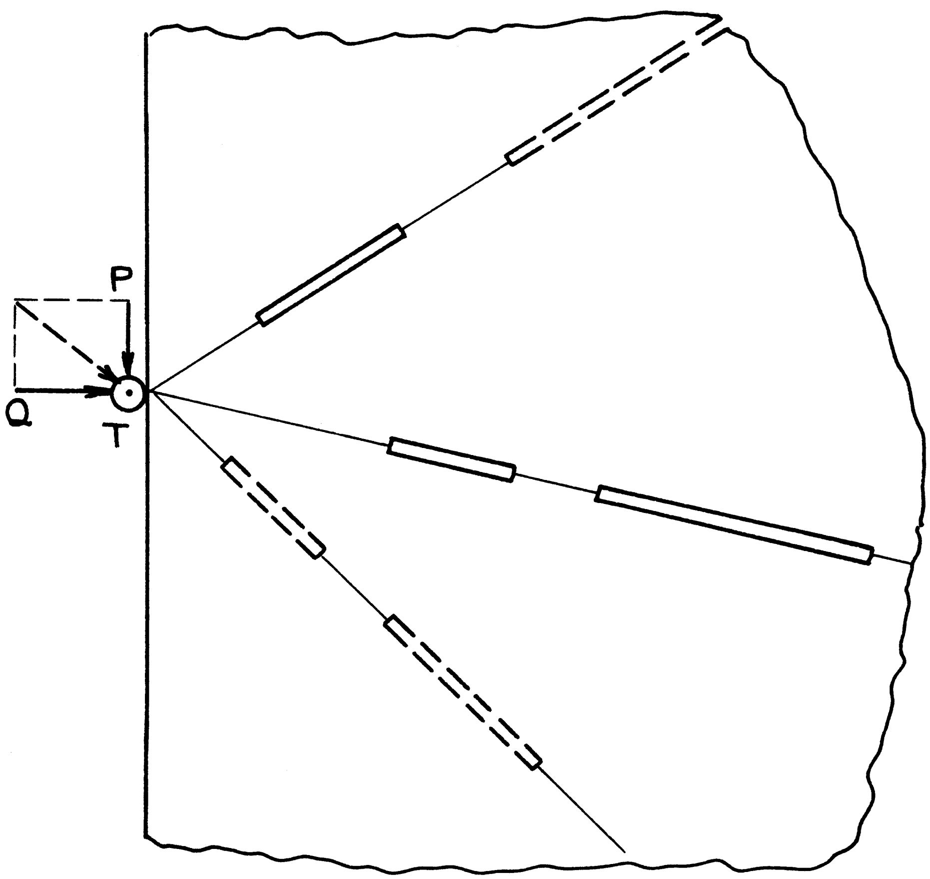
Accuracy: K_I, K_{II} 1%
K_{III} Exact

References: K_I **Isida 1965a, 1970a**; K_{II} **Tada 1973**; K_{III} **Erdogan 1962; Sih 1964**



$$K_I = K_{II} = K_{III} = 0$$

Method: Special case of **page 10.2a**
Accuracy: Exact
Reference: **Tada 1973**

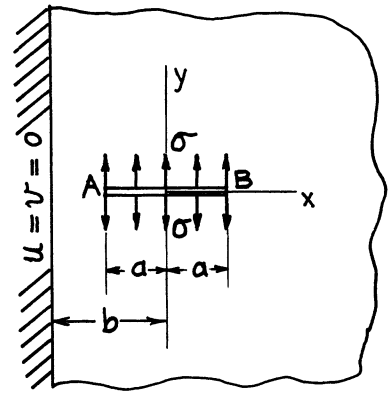


$$K_I = K_{II} = K_{III} = 0$$

Methods: Westergaard Stress Function, etc. (Simple Radial Stresses for P and Q)
Accuracy: Exact
Reference: **Tada 1985**

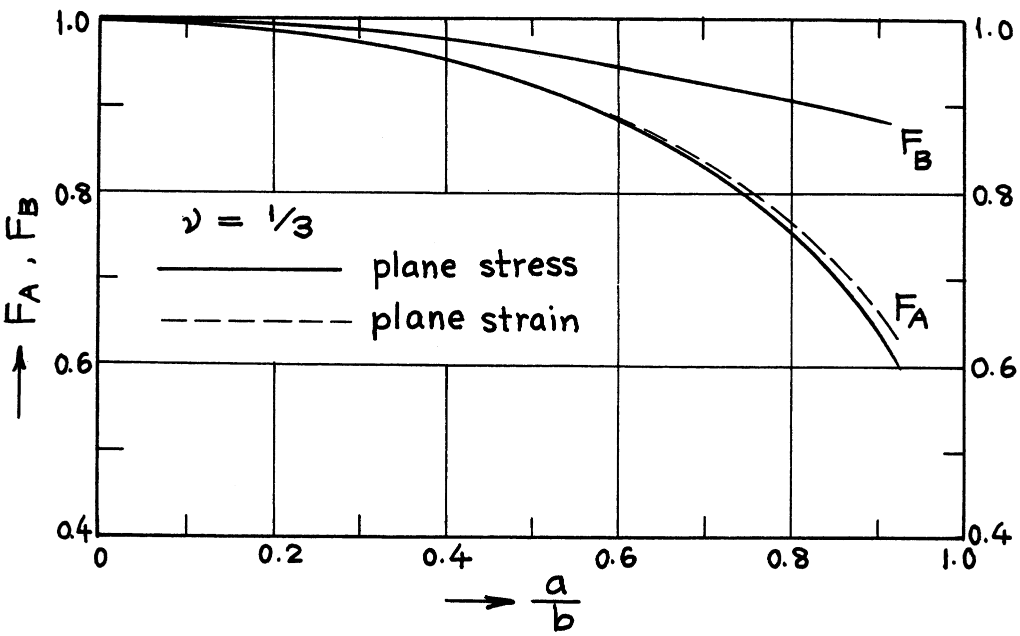
$$K_{I,A} = \sigma\sqrt{\pi a} \cdot F_A\left({}^{a}\!/_{b}\right)$$

$$K_{I,B} = \sigma\sqrt{\pi a} \cdot F_B\left({}^{a}\!/_{b}\right)$$



$$F_A\left({}^{a}\!/_{b}\right) = 1 - .175\left({}^{a}\!/_{b}\right)^2 - .245\left({}^{a}\!/_{b}\right)^3 \quad \left({}^{a}\!/_{b} \le .8\right)$$

$$F_B\left({}^{a}\!/_{b}\right) = 1 - .145\left({}^{a}\!/_{b}\right)^2 \quad \left({}^{a}\!/_{b} \le .9\right)$$



Method: Series Expansions of Complex Potentials
Accuracy: Curves are based on numerical values with 0.1% accuracy
Formulas: F_A 1% for ${}^{a}\!/_{b} < 0.8$; F_B 1% for ${}^{a}\!/_{b} < 0.9$
References: **Isida 1970a; Tada 1985**